ImgFilt

1.0

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1 Namespace Index 1
1.1 Namespace List . . . 1

2 Hierarchical Index 3
2.1 Class Hierarchy . . . 3

3 Class Index 5
3.1 Class List . . . 5

4 File Index 7
4.1 File List . . . 7

5 Namespace Documentation 9
5.1 kernelGenerators Namespace Reference . . . 9
5.1.1 Function Documentation . . . 9
5.1.1.1 createBoxBlurKernel() . . . 9
5.1.1.2 createGaussianBlurKernel() . . . 9

6 Class Documentation 11
6.1 BatchProcessor Class Reference . . . 11
6.1.1 Constructor & Destructor Documentation . . . 11
6.1.1.1 BatchProcessor() . . . 11
6.1.1.2 ∼BatchProcessor() . . . 11
6.1.2 Member Function Documentation . . . 12
6.1.2.1 handleCommand() . . . 12
6.1.2.2 process() . . . 12
6.2 BITMAPFILEHEADER Struct Reference . . . 13
6.2.1 Member Data Documentation . . . 13
6.2.1.1 fileSize . . . 13
6.2.1.2 fileType . . . 13
6.2.1.3 offsetData . . . 13
6.2.1.4 reserved1 . . . 13
6.2.1.5 reserved2 . . . 13
6.3 BITMAPINFOHEADER Struct Reference . . . 14
6.3.1 Member Data Documentation . . . 14
6.3.1.1 bitCount . . . 14
6.3.1.2 colorsUsed . . . 14
6.3.1.3 compression . . . 14
6.3.1.4 height . . . 14
6.3.1.5 importantColors . . . 14
6.3.1.6 planes . . . 14
6.3.1.7 size . . . 15
6.3.1.8 sizeImage . . . 15
6.3.1.9 width . . . 15

6.3.1.10 xPixelsPerMeter . . . 15
6.3.1.11 yPixelsPerMeter . . . 15
6.4 BMPImageHandler Class Reference . . . 15
6.4.1 Constructor & Destructor Documentation . . . 16
6.4.1.1 BMPImageHandler() . . . 16
6.4.1.2 ∼BMPImageHandler() . . . 16
6.4.2 Member Function Documentation . . . 17
6.4.2.1 flipVertical() . . . 17
6.4.2.2 load() . . . 17
6.4.2.3 save() . . . 18
6.5 ContrastBrightnessFilter Class Reference . . . 18
6.5.1 Constructor & Destructor Documentation . . . 19
6.5.1.1 ContrastBrightnessFilter() . . . 19
6.5.1.2 ∼ContrastBrightnessFilter() . . . 20
6.5.2 Member Function Documentation . . . 20
6.5.2.1 apply() . . . 20
6.6 ConvolutionFilter Class Reference . . . 21
6.6.1 Constructor & Destructor Documentation . . . 21
6.6.1.1 ConvolutionFilter() . . . 21
6.6.1.2 ∼ConvolutionFilter() . . . 22
6.6.2 Member Function Documentation . . . 22
6.6.2.1 apply() . . . 22
6.7 FilterFactory Class Reference . . . 22
6.7.1 Member Function Documentation . . . 23
6.7.1.1 buildFilter() . . . 23
6.8 FilterPipeline Class Reference . . . 23
6.8.1 Constructor & Destructor Documentation . . . 24
6.8.1.1 FilterPipeline() . . . 24
6.8.1.2 ∼FilterPipeline() . . . 24
6.8.2 Member Function Documentation . . . 24
6.8.2.1 add() . . . 24
6.8.2.2 apply() . . . 24
6.8.2.3 clear() . . . 24
6.9 FilterSpec Class Reference . . . 25
6.9.1 Constructor & Destructor Documentation . . . 25
6.9.1.1 FilterSpec() . . . 25
6.9.1.2 ∼FilterSpec() . . . 25
6.9.2 Member Function Documentation . . . 26
6.9.2.1 clear() . . . 26
6.9.3 Member Data Documentation . . . 26
6.9.3.1 blur . . . 26
6.9.3.2 bOffset . . . 26

6.9.3.3 brightnessFactor . . . 26
6.9.3.4 contrastFactor . . . 26
6.9.3.5 gaussianBlurSigma . . . 26
6.9.3.6 gaussianBlurSize . . . 26
6.9.3.7 gOffset . . . 26
6.9.3.8 hasBlur . . . 27
6.9.3.9 hasBrightness . . . 27
6.9.3.10 hasContrast . . . 27
6.9.3.11 hasGaussian . . . 27
6.9.3.12 hasMedian . . . 27
6.9.3.13 hasRGBOffset . . . 27
6.9.3.14 medianKernelSize . . . 27
6.9.3.15 rOffset . . . 27
6.10 IFilter Class Reference . . . 28
6.10.1 Constructor & Destructor Documentation . . . 28
6.10.1.1 ~IFilter() . . . 28
6.10.2 Member Function Documentation . . . 28
6.10.2.1 apply() . . . 28
6.11 IImageHandler Class Reference . . . 29
6.11.1 Constructor & Destructor Documentation . . . 29
6.11.1.1 ~IImageHandler() . . . 29
6.11.2 Member Function Documentation . . . 29
6.11.2.1 load() . . . 29
6.11.2.2 save() . . . 29
6.12 Image Class Reference . . . 30
6.12.1 Constructor & Destructor Documentation . . . 30
6.12.1.1 Image() [1/2] . . . 30
6.12.1.2 Image() [2/2] . . . 30
6.12.2 Member Function Documentation . . . 30
6.12.2.1 at() [1/2] . . . 30
6.12.2.2 at() [2/2] . . . 31
6.12.2.3 getHeight() . . . 31
6.12.2.4 getWidth() . . . 32
6.12.2.5 resize() . . . 32
6.12.2.6 setPixel() . . . 32
6.13 Kernel Class Reference . . . 33
6.13.1 Constructor & Destructor Documentation . . . 33
6.13.1.1 Kernel() [1/2] . . . 33
6.13.1.2 Kernel() [2/2] . . . 34
6.13.2 Member Function Documentation . . . 34
6.13.2.1 absToRel() . . . 34
6.13.2.2 centerX() . . . 34

6.13.2.3 centerY() . . . 35
6.13.2.4 cols() . . . 35
6.13.2.5 getValue() . . . 35
6.13.2.6 getValues() [1/2] . . . 36
6.13.2.7 getValues() [2/2] . . . 36
6.13.2.8 isNormalized() . . . 36
6.13.2.9 normalize() . . . 36
6.13.2.10 rows() . . . 37
6.13.2.11 scale() . . . 37
6.13.2.12 set() [1/2] . . . 38
6.13.2.13 set() [2/2] . . . 38
6.13.2.14 sum() . . . 38
6.14 MedianFilter Class Reference . . . 39
6.14.1 Constructor & Destructor Documentation . . . 40
6.14.1.1 MedianFilter() . . . 40
6.14.1.2 ∼MedianFilter() . . . 40
6.14.2 Member Function Documentation . . . 41
6.14.2.1 absToRel() . . . 41
6.14.2.2 apply() . . . 41
6.15 Pixel Class Reference . . . 42
6.15.1 Constructor & Destructor Documentation . . . 43
6.15.1.1 Pixel() . . . 43
6.15.2 Member Function Documentation . . . 43
6.15.2.1 getA() . . . 43
6.15.2.2 getB() . . . 44
6.15.2.3 getG() . . . 44
6.15.2.4 getR() . . . 44
6.15.2.5 operator∗() . . . 45
6.15.2.6 operator+=() . . . 45
6.15.2.7 setA() . . . 45
6.15.2.8 setB() . . . 46
6.15.2.9 setG() . . . 46
6.15.2.10 setPixel() . . . 46
6.15.2.11 setR() . . . 47
6.15.2.12 setRGBA() . . . 47
6.16 RGBOffsetFilter Class Reference . . . 48
6.16.1 Constructor & Destructor Documentation . . . 48
6.16.1.1 RGBOffsetFilter() . . . 48
6.16.1.2 ∼RGBOffsetFilter() . . . 49
6.16.2 Member Function Documentation . . . 49
6.16.2.1 apply() . . . 49

7 File Documentation 51
7.1 include/filt_op/FilterFactory.hpp File Reference . . . 51
7.2 FilterFactory.hpp . . . 52
7.3 include/filt_op/FilterPipeline.hpp File Reference . . . 52
7.4 FilterPipeline.hpp . . . 53
7.5 include/filt_op/IFilter.hpp File Reference . . . 54
7.6 IFilter.hpp . . . 55
7.7 include/filt_op/kernelGenerators.hpp File Reference . . . 55
7.8 kernelGenerators.hpp . . . 56
7.9 include/filters/ContrastBrightnessFilter.hpp File Reference . . . 56
7.10 ContrastBrightnessFilter.hpp . . . 57
7.11 include/filters/ConvolutionFilter.hpp File Reference . . . 57
7.12 ConvolutionFilter.hpp . . . 58
7.13 include/filters/MedianFilter.hpp File Reference . . . 59
7.14 MedianFilter.hpp . . . 60
7.15 include/filters/RGBOffsetFilter.hpp File Reference . . . 60
7.16 RGBOffsetFilter.hpp . . . 61
7.17 include/io/BatchProcessor.hpp File Reference . . . 61
7.18 BatchProcessor.hpp . . . 62
7.19 include/io/BMPImageHandler.hpp File Reference . . . 63
7.20 BMPImageHandler.hpp . . . 64
7.21 include/io/IImageHandler.hpp File Reference . . . 64
7.22 IImageHandler.hpp . . . 65
7.23 include/types/CommandType.hpp File Reference . . . 65
7.23.1 Enumeration Type Documentation . . . 66
7.23.1.1 CommandType . . . 66
7.24 CommandType.hpp . . . 66
7.25 include/types/FilterSpec.hpp File Reference . . . 67
7.26 FilterSpec.hpp . . . 67
7.27 include/types/Image.hpp File Reference . . . 68
7.28 Image.hpp . . . 69
7.29 include/types/Kernel.hpp File Reference . . . 69
7.29.1 Typedef Documentation . . . 70
7.29.1.1 KernelMatrix . . . 70
7.30 Kernel.hpp . . . 70
7.31 include/types/microsoft.hpp File Reference . . . 71
7.32 microsoft.hpp . . . 71
7.33 include/types/Pixel.hpp File Reference . . . 72
7.34 Pixel.hpp . . . 72
7.35 src/filt_op/FilterFactory.cpp File Reference . . . 73
7.36 src/filt_op/FilterPipeline.cpp File Reference . . . 74
7.37 src/filt_op/kernelGenerators.cpp File Reference . . . 74

7.38 src/filters/ContrastBrightnessFilter.cpp File Reference 75
7.39 src/filters/ConvolutionFilter.cpp File Reference 76
7.40 src/filters/MedianFilter.cpp File Reference 77
7.41 src/filters/RGBOffsetFilter.cpp File Reference 78
7.42 src/io/BatchProcessor.cpp File Reference 79
7.43 src/io/BMPImageHandler.cpp File Reference 80
7.44 src/main.cpp File Reference 80
7.44.1 Function Documentation 81
7.44.1.1 main() 81
7.45 src/types/Kernel.cpp File Reference 81

Index **83**

Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

kernelGenerators 9

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

BatchProcessor 11
BITMAPFILEHEADER 13
BITMAPINFOHEADER 14
FilterFactory 22
FilterPipeline 23
FilterSpec 25
IFilter 28
    ContrastBrightnessFilter 18
    ConvolutionFilter 21
    MedianFilter 39
    RGBOffsetFilter 48
IImageHandler 29
    BMPImageHandler 15
Image 30
Kernel 33
Pixel 42

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

BatchProcessor 11
BITMAPFILEHEADER 13
BITMAPINFOHEADER 14
BMPImageHandler 15
ContrastBrightnessFilter 18
ConvolutionFilter 21
FilterFactory 22
FilterPipeline 23
FilterSpec 25
IFilter 28
IImageHandler 29
Image 30
Kernel 33
MedianFilter 39
Pixel 42
RGBOffsetFilter 48

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

include/filt_op/FilterFactory.hpp . . . 51
include/filt_op/FilterPipeline.hpp . . . 52
include/filt_op/IFilter.hpp . . . 54
include/filt_op/kernelGenerators.hpp . . . 55
include/filters/ContrastBrightnessFilter.hpp . . . 56
include/filters/ConvolutionFilter.hpp . . . 57
include/filters/MedianFilter.hpp . . . 59
include/filters/RGBOffsetFilter.hpp . . . 60
include/io/BatchProcessor.hpp . . . 61
include/io/BMPImageHandler.hpp . . . 63
include/io/IImageHandler.hpp . . . 64
include/types/CommandType.hpp . . . 65
include/types/FilterSpec.hpp . . . 67
include/types/Image.hpp . . . 68
include/types/Kernel.hpp . . . 69
include/types/microsoft.hpp . . . 71
include/types/Pixel.hpp . . . 72
src/main.cpp . . . 80
src/filt_op/FilterFactory.cpp . . . 73
src/filt_op/FilterPipeline.cpp . . . 74
src/filt_op/kernelGenerators.cpp . . . 74
src/filters/ContrastBrightnessFilter.cpp . . . 75
src/filters/ConvolutionFilter.cpp . . . 76
src/filters/MedianFilter.cpp . . . 77
src/filters/RGBOffsetFilter.cpp . . . 78
src/io/BatchProcessor.cpp . . . 79
src/io/BMPImageHandler.cpp . . . 80
src/types/Kernel.cpp . . . 81

Chapter 5

Namespace Documentation

5.1 kernelGenerators Namespace Reference

Functions

- Kernel createBoxBlurKernel (int size)
- Kernel createGaussianBlurKernel (int size, double sigma)

5.1.1 Function Documentation

5.1.1.1 createBoxBlurKernel()

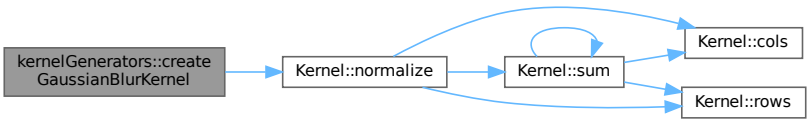
```
Kernel kernelGenerators::createBoxBlurKernel (
            int size )
```

Here is the caller graph for this function:

5.1.1.2 createGaussianBlurKernel()

```
Kernel kernelGenerators::createGaussianBlurKernel (
            int size,
            double sigma )
```

Here is the call graph for this function:



Here is the caller graph for this function:

Chapter 6

Class Documentation

6.1 BatchProcessor Class Reference

```
#include <BatchProcessor.hpp>
```

Public Member Functions

- BatchProcessor ()=default
- ~BatchProcessor ()=default
- void process (const std::string &filename)
- void handleCommand (const std::string &command, std::istringstream &args)

6.1.1 Constructor & Destructor Documentation

6.1.1.1 BatchProcessor()

```
BatchProcessor::BatchProcessor ( ) [default]
```

6.1.1.2 ~BatchProcessor()

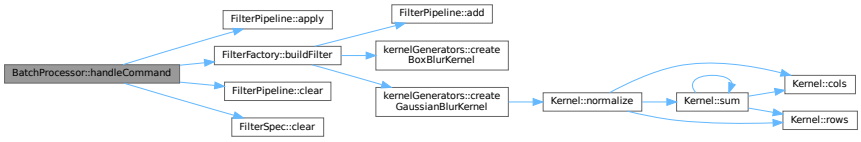
```
BatchProcessor::~BatchProcessor ( ) [default]
```

6.1.2 Member Function Documentation

6.1.2.1 handleCommand()

```
void BatchProcessor::handleCommand (
            const std::string & command,
            std::istringstream & args )
```

Here is the call graph for this function:

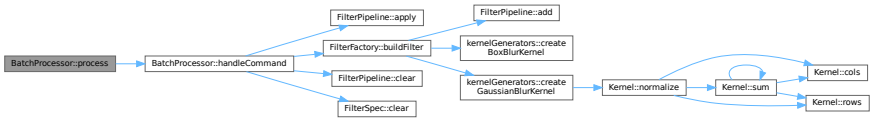


Here is the caller graph for this function:

6.1.2.2 process()

```
void BatchProcessor::process (
            const std::string & filename )
```

Here is the call graph for this function:



Here is the caller graph for this function:

The documentation for this class was generated from the following files:

- include/io/BatchProcessor.hpp
- src/io/BatchProcessor.cpp

6.2 BITMAPFILEHEADER Struct Reference

```
#include <microsoft.hpp>
```

Public Attributes

- uint16_t fileType
- uint32_t fileSize
- uint16_t reserved1
- uint16_t reserved2
- uint32_t offsetData

6.2.1 Member Data Documentation

6.2.1.1 fileSize

```
uint32_t BITMAPFILEHEADER::fileSize
```

6.2.1.2 fileType

```
uint16_t BITMAPFILEHEADER::fileType
```

6.2.1.3 offsetData

```
uint32_t BITMAPFILEHEADER::offsetData
```

6.2.1.4 reserved1

```
uint16_t BITMAPFILEHEADER::reserved1
```

6.2.1.5 reserved2

```
uint16_t BITMAPFILEHEADER::reserved2
```

The documentation for this struct was generated from the following file:

- include/types/microsoft.hpp

6.3 BITMAPINFOHEADER Struct Reference

```
#include <microsoft.hpp>
```

Public Attributes

- uint32_t size
- int32_t width
- int32_t height
- uint16_t planes
- uint16_t bitCount
- uint32_t compression
- uint32_t sizeImage
- int32_t xPixelsPerMeter
- int32_t yPixelsPerMeter
- uint32_t colorsUsed
- uint32_t importantColors

6.3.1 Member Data Documentation

6.3.1.1 bitCount

```
uint16_t BITMAPINFOHEADER::bitCount
```

6.3.1.2 colorsUsed

```
uint32_t BITMAPINFOHEADER::colorsUsed
```

6.3.1.3 compression

```
uint32_t BITMAPINFOHEADER::compression
```

6.3.1.4 height

```
int32_t BITMAPINFOHEADER::height
```

6.3.1.5 importantColors

```
uint32_t BITMAPINFOHEADER::importantColors
```

6.3.1.6 planes

```
uint16_t BITMAPINFOHEADER::planes
```

6.3.1.7 size

```
uint32_t BITMAPINFOHEADER::size
```

6.3.1.8 sizeImage

```
uint32_t BITMAPINFOHEADER::sizeImage
```

6.3.1.9 width

```
int32_t BITMAPINFOHEADER::width
```

6.3.1.10 xPixelsPerMeter

```
int32_t BITMAPINFOHEADER::xPixelsPerMeter
```

6.3.1.11 yPixelsPerMeter

```
int32_t BITMAPINFOHEADER::yPixelsPerMeter
```

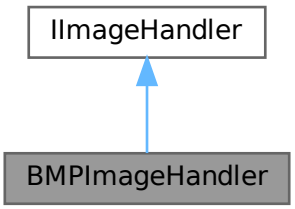
The documentation for this struct was generated from the following file:

- include/types/microsoft.hpp

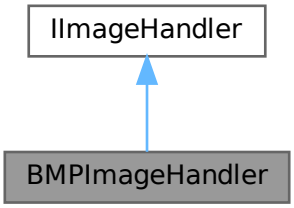
6.4 BMPImageHandler Class Reference

```
#include <BMPImageHandler.hpp>
```

Inheritance diagram for BMPImageHandler:



Collaboration diagram for BMPImageHandler:



Public Member Functions

- BMPImageHandler ()=default
- ~BMPImageHandler () override
- std::unique_ptr< Image > load (const std::string &filename) const override
- void save (const std::string &filename, const Image &image) const override
- void flipVertical (Image &img, BITMAPINFOHEADER &header) const

Public Member Functions inherited from IImageHandler

- virtual ~IImageHandler ()=default

6.4.1 Constructor & Destructor Documentation

6.4.1.1 BMPImageHandler()

```
BMPImageHandler::BMPImageHandler ( ) [default]
```

6.4.1.2 ~BMPImageHandler()

```
BMPImageHandler::~BMPImageHandler ( ) [override], [default]
```

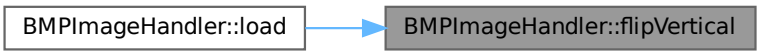
6.4.2 Member Function Documentation

6.4.2.1 flipVertical()

```
void BMPImageHandler::flipVertical (
            Image & img,
            BITMAPINFOHEADER & header ) const
```

Here is the call graph for this function:

Here is the caller graph for this function:



6.4.2.2 load()

```
std::unique_ptr< Image > BMPImageHandler::load (
            const std::string & filename ) const  [override], [virtual]
```

Implements IImageHandler.

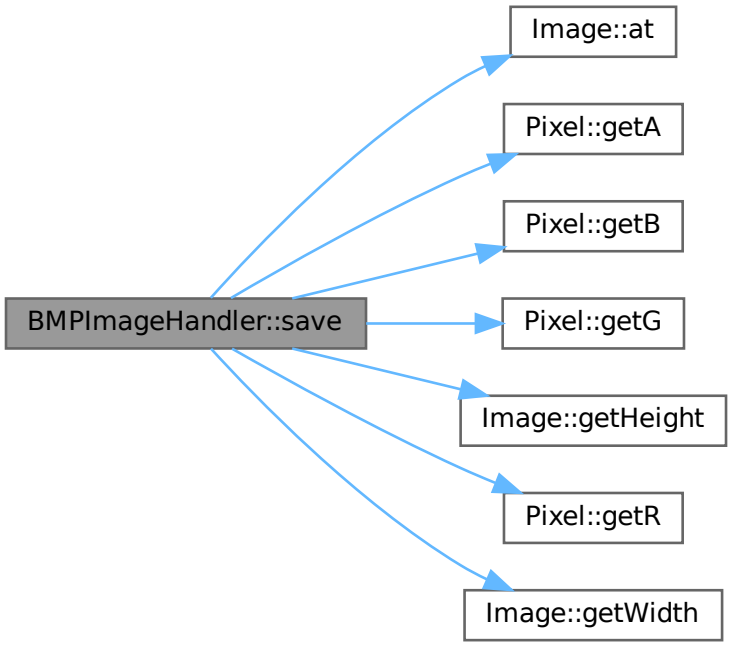
Here is the call graph for this function:

6.4.2.3 save()

```
void BMPImageHandler::save (
            const std::string & filename,
            const Image & image ) const  [override], [virtual]
```

Implements IImageHandler.

Here is the call graph for this function:



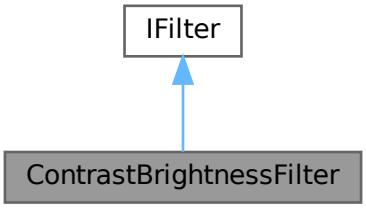
The documentation for this class was generated from the following files:

- include/io/BMPImageHandler.hpp
- src/io/BMPImageHandler.cpp

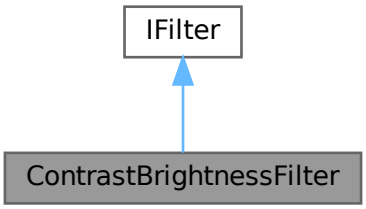
6.5 ContrastBrightnessFilter Class Reference

```
#include <ContrastBrightnessFilter.hpp>
```

Inheritance diagram for ContrastBrightnessFilter:



Collaboration diagram for ContrastBrightnessFilter:



Public Member Functions

- ContrastBrightnessFilter (double contrastFactor=1.0, double brightnessFactor=0.0)
- void apply (Image &img) const override
- ~ContrastBrightnessFilter ()=default

Public Member Functions inherited from IFilter

- virtual ~IFilter ()=default

6.5.1 Constructor & Destructor Documentation

6.5.1.1 ContrastBrightnessFilter()

```
ContrastBrightnessFilter::ContrastBrightnessFilter (
            double contrastFactor = 1.0,
            double brightnessFactor = 0.0 )  [inline]
```

6.5.1.2 ~ContrastBrightnessFilter()

```
ContrastBrightnessFilter::~ContrastBrightnessFilter ( ) [default]
```

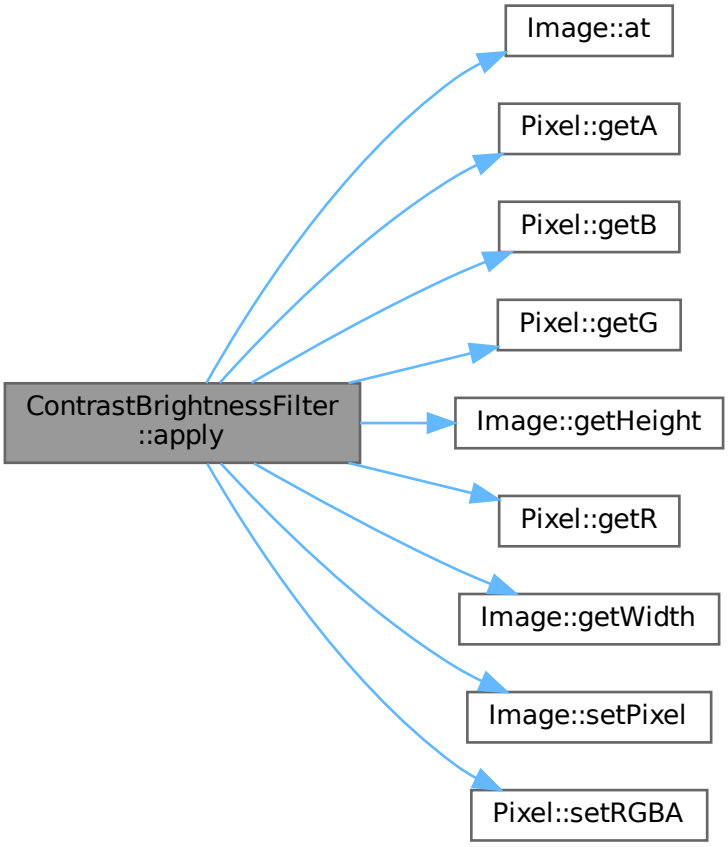
6.5.2 Member Function Documentation

6.5.2.1 apply()

```
void ContrastBrightnessFilter::apply (
            Image & img ) const [override], [virtual]
```

Implements IFilter.

Here is the call graph for this function:



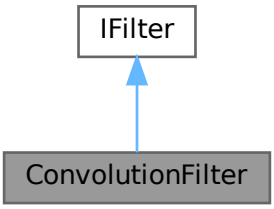
The documentation for this class was generated from the following files:

- include/filters/ContrastBrightnessFilter.hpp
- src/filters/ContrastBrightnessFilter.cpp

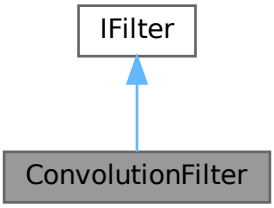
6.6 ConvolutionFilter Class Reference

```
#include <ConvolutionFilter.hpp>
```

Inheritance diagram for ConvolutionFilter:



Collaboration diagram for ConvolutionFilter:



Public Member Functions

- ConvolutionFilter (const Kernel &kernel)
- ~ConvolutionFilter ()=default
- void apply (Image &img) const override

Public Member Functions inherited from IFilter

- virtual ~IFilter ()=default

6.6.1 Constructor & Destructor Documentation

6.6.1.1 ConvolutionFilter()

```
ConvolutionFilter::ConvolutionFilter (
            const Kernel & kernel )  [inline]
```

6.6.1.2 ~ConvolutionFilter()

```
ConvolutionFilter::~ConvolutionFilter ( )  [default]
```

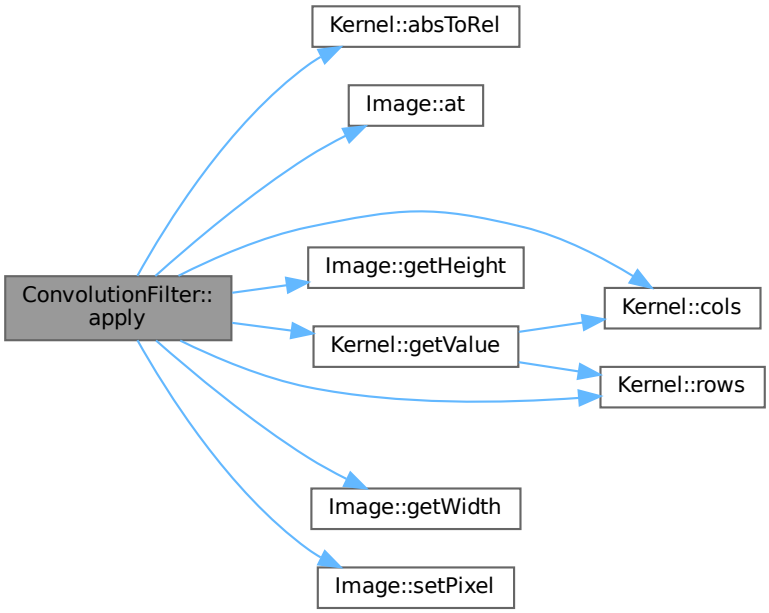
6.6.2 Member Function Documentation

6.6.2.1 apply()

```
void ConvolutionFilter::apply (
            Image & img ) const  [override], [virtual]
```

Implements IFilter.

Here is the call graph for this function:



The documentation for this class was generated from the following files:

- include/filters/ConvolutionFilter.hpp
- src/filters/ConvolutionFilter.cpp

6.7 FilterFactory Class Reference

```
#include <FilterFactory.hpp>
```

Static Public Member Functions

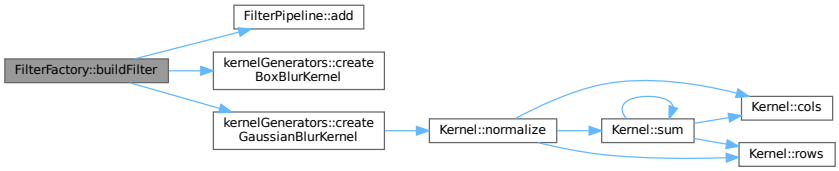
- static void buildFilter (FilterPipeline &pipeline, const FilterSpec &spec)

6.7.1 Member Function Documentation

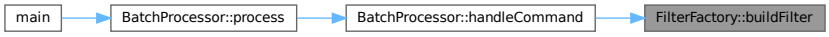
6.7.1.1 buildFilter()

```
void FilterFactory::buildFilter (
            FilterPipeline & pipeline,
            const FilterSpec & spec )  [static]
```

Here is the call graph for this function:



Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- include/filt_op/FilterFactory.hpp
- src/filt_op/FilterFactory.cpp

6.8 FilterPipeline Class Reference

```
#include <FilterPipeline.hpp>
```

Public Member Functions

- FilterPipeline ()=default
- ~FilterPipeline ()=default
- void apply (Image &img)
- void add (std::unique_ptr< IFilter > filter)
- void clear ()

6.8.1 Constructor & Destructor Documentation

6.8.1.1 FilterPipeline()

```
FilterPipeline::FilterPipeline ( ) [default]
```

6.8.1.2 ~FilterPipeline()

```
FilterPipeline::~FilterPipeline ( ) [default]
```

6.8.2 Member Function Documentation

6.8.2.1 add()

```
void FilterPipeline::add (
            std::unique_ptr< IFilter > filter )
```

Here is the caller graph for this function:

6.8.2.2 apply()

```
void FilterPipeline::apply (
            Image & img )
```

Here is the caller graph for this function:

6.8.2.3 clear()

```
void FilterPipeline::clear ( )
```

Here is the caller graph for this function:

The documentation for this class was generated from the following files:

- include/filt_op/FilterPipeline.hpp
- src/filt_op/FilterPipeline.cpp

6.9 FilterSpec Class Reference

```
#include <FilterSpec.hpp>
```

Public Member Functions

- FilterSpec ()
- ~FilterSpec ()=default
- void clear ()

Public Attributes

- bool hasContrast = false
- double contrastFactor = 0
- bool hasBrightness = false
- double brightnessFactor = 0
- bool hasBlur = false
- unsigned blur = 0
- bool hasGaussian = false
- unsigned gaussianBlurSize = 0
- double gaussianBlurSigma = 1.0
- bool hasMedian = false
- uint medianKernelSize = 0
- bool hasRGBOffset = false
- double rOffset = 0
- double gOffset = 0
- double bOffset = 0

6.9.1 Constructor & Destructor Documentation

6.9.1.1 FilterSpec()

```
FilterSpec::FilterSpec ( ) [inline]
```

Here is the call graph for this function:

6.9.1.2 ~FilterSpec()

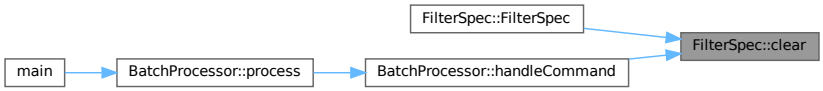
```
FilterSpec::~FilterSpec ( ) [default]
```

6.9.2 Member Function Documentation

6.9.2.1 clear()

```
void FilterSpec::clear ( ) [inline]
```

Here is the caller graph for this function:



6.9.3 Member Data Documentation

6.9.3.1 blur

```
unsigned FilterSpec::blur = 0
```

6.9.3.2 bOffset

```
double FilterSpec::bOffset = 0
```

6.9.3.3 brightnessFactor

```
double FilterSpec::brightnessFactor = 0
```

6.9.3.4 contrastFactor

```
double FilterSpec::contrastFactor = 0
```

6.9.3.5 gaussianBlurSigma

```
double FilterSpec::gaussianBlurSigma = 1.0
```

6.9.3.6 gaussianBlurSize

```
unsigned FilterSpec::gaussianBlurSize = 0
```

6.9.3.7 gOffset

```
double FilterSpec::gOffset = 0
```

6.9.3.8 hasBlur

```
bool FilterSpec::hasBlur = false
```

6.9.3.9 hasBrightness

```
bool FilterSpec::hasBrightness = false
```

6.9.3.10 hasContrast

```
bool FilterSpec::hasContrast = false
```

6.9.3.11 hasGaussian

```
bool FilterSpec::hasGaussian = false
```

6.9.3.12 hasMedian

```
bool FilterSpec::hasMedian = false
```

6.9.3.13 hasRGBOffset

```
bool FilterSpec::hasRGBOffset = false
```

6.9.3.14 medianKernelSize

```
uint FilterSpec::medianKernelSize = 0
```

6.9.3.15 rOffset

```
double FilterSpec::rOffset = 0
```

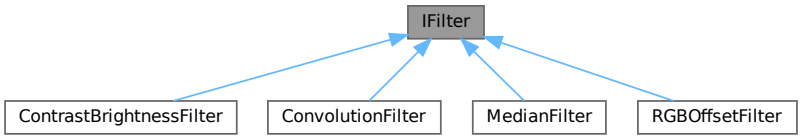
The documentation for this class was generated from the following file:

- include/types/FilterSpec.hpp

6.10 IFilter Class Reference

```
#include <IFilter.hpp>
```

Inheritance diagram for IFilter:



Public Member Functions

- virtual ~IFilter ()=default
- virtual void apply (Image &img) const =0

6.10.1 Constructor & Destructor Documentation

6.10.1.1 ~IFilter()

```
virtual IFilter::~IFilter ( )  [virtual], [default]
```

6.10.2 Member Function Documentation

6.10.2.1 apply()

```
virtual void IFilter::apply (
            Image & img ) const  [pure virtual]
```

Implemented in ContrastBrightnessFilter, ConvolutionFilter, MedianFilter, and RGBOffsetFilter.

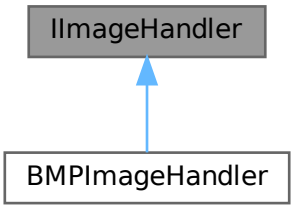
The documentation for this class was generated from the following file:

- include/filt_op/IFilter.hpp

6.11 IImageHandler Class Reference

```
#include <IImageHandler.hpp>
```

Inheritance diagram for IImageHandler:



Public Member Functions

- virtual ~IImageHandler ()=default
- virtual std::unique_ptr< Image > load (const std::string &filename) const =0
- virtual void save (const std::string &filename, const Image &image) const =0

6.11.1 Constructor & Destructor Documentation

6.11.1.1 ~IImageHandler()

```
virtual IImageHandler::~IImageHandler ( )  [virtual], [default]
```

6.11.2 Member Function Documentation

6.11.2.1 load()

```
virtual std::unique_ptr< Image > IImageHandler::load (
            const std::string & filename ) const  [pure virtual]
```

Implemented in BMPImageHandler.

6.11.2.2 save()

```
virtual void IImageHandler::save (
            const std::string & filename,
            const Image & image ) const  [pure virtual]
```

Implemented in BMPImageHandler.

The documentation for this class was generated from the following file:

- include/io/IImageHandler.hpp

6.12 Image Class Reference

```
#include <Image.hpp>
```

Public Member Functions

- Image ()=default
- Image (int width, int height)
- const Pixel & at (int row, int col) const
- Pixel & at (int row, int col)
- void resize (int width, int height)
- void setPixel (int row, int col, const Pixel &pixel)
- int getWidth () const
- int getHeight () const

6.12.1 Constructor & Destructor Documentation

6.12.1.1 Image() [1/2]

```
Image::Image ( ) [default]
```

6.12.1.2 Image() [2/2]

```
Image::Image (
            int width,
            int height ) [inline]
```

6.12.2 Member Function Documentation

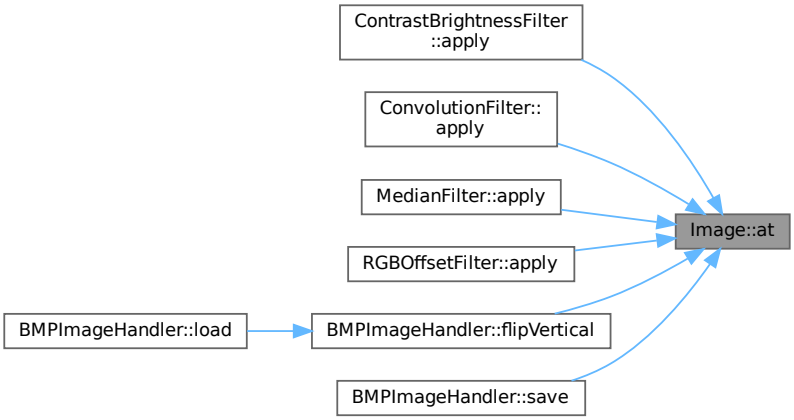
6.12.2.1 at() [1/2]

```
Pixel & Image::at (
            int row,
            int col ) [inline]
```

6.12.2.2 at() [2/2]

```
const Pixel & Image::at (
            int row,
            int col ) const  [inline]
```

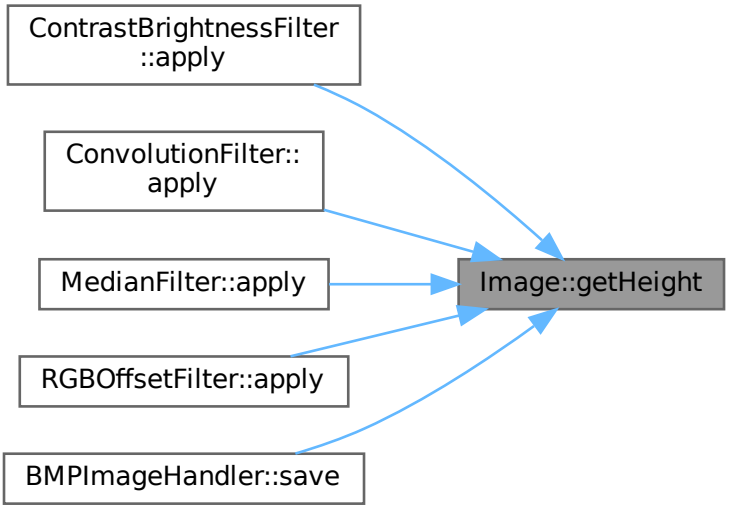
Here is the caller graph for this function:



6.12.2.3 getHeight()

```
int Image::getHeight ( ) const  [inline]
```

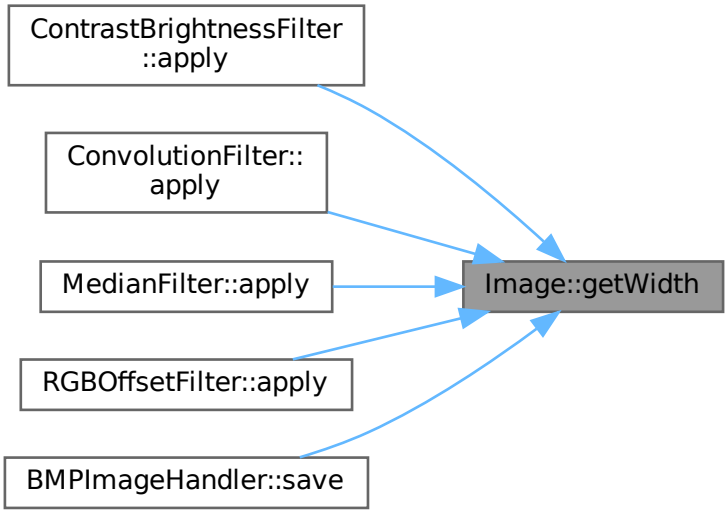
Here is the caller graph for this function:



6.12.2.4 getWidth()

```
int Image::getWidth ( ) const  [inline]
```

Here is the caller graph for this function:



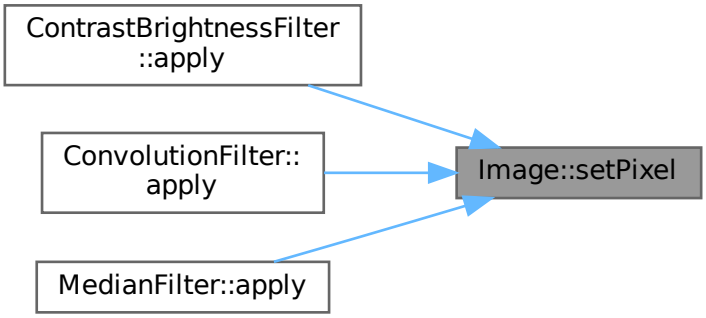
6.12.2.5 resize()

```
void Image::resize (
            int width,
            int height )  [inline]
```

6.12.2.6 setPixel()

```
void Image::setPixel (
            int row,
            int col,
            const Pixel & pixel )  [inline]
```

Here is the caller graph for this function:



The documentation for this class was generated from the following file:

- include/types/Image.hpp

6.13 Kernel Class Reference

```
#include <Kernel.hpp>
```

Public Member Functions

- Kernel ()=default
- Kernel (KernelMatrix values)
- double getValue (int row, int col) const
- void set (int row, int col, double value)
- void set (const KernelMatrix &values)
- const KernelMatrix & getValues () const
- KernelMatrix & getValues ()
- int centerX () const
- int centerY () const
- int rows () const
- int cols () const
- int absToRel (int coordinate, int size) const
- double sum () const
- void normalize (double epsilon=1e-6)
- bool isNormalized (double epsilon=1e-6) const
- void scale (double factor)

6.13.1 Constructor & Destructor Documentation

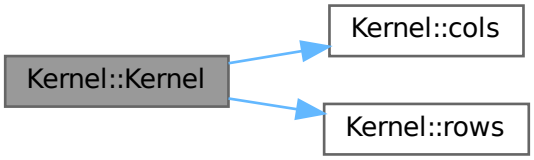
6.13.1.1 Kernel() [1/2]

```
Kernel::Kernel ( ) [default]
```

6.13.1.2 Kernel() [2/2]

```
Kernel::Kernel (
            KernelMatrix values )
```

Here is the call graph for this function:

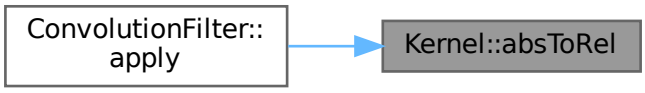


6.13.2 Member Function Documentation

6.13.2.1 absToRel()

```
int Kernel::absToRel (
            int coordinate,
            int size ) const
```

Here is the caller graph for this function:



6.13.2.2 centerX()

```
int Kernel::centerX ( ) const
```

Here is the call graph for this function:

6.13.2.3 centerY()

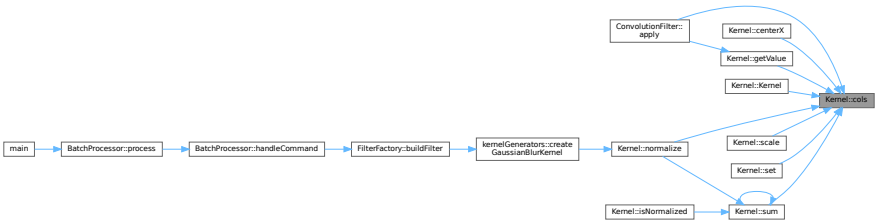
```
int Kernel::centerY ( ) const
```

Here is the call graph for this function:

6.13.2.4 cols()

```
int Kernel::cols ( ) const
```

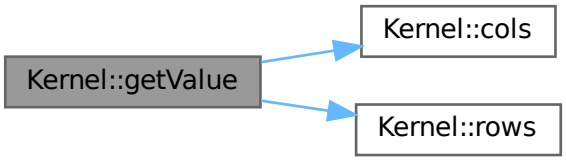
Here is the caller graph for this function:



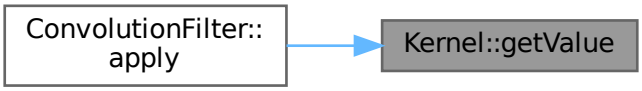
6.13.2.5 getValue()

```
double Kernel::getValue (
            int row,
            int col ) const
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.13.2.6 getValues() [1/2]

```
KernelMatrix & Kernel::getValues ( )
```

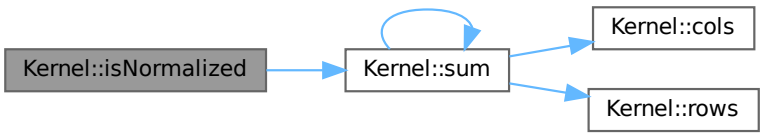
6.13.2.7 getValues() [2/2]

```
const KernelMatrix & Kernel::getValues ( ) const
```

6.13.2.8 isNormalized()

```
bool Kernel::isNormalized (
            double epsilon = 1e-6 ) const
```

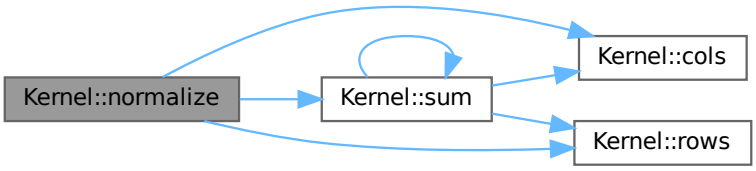
Here is the call graph for this function:



6.13.2.9 normalize()

```
void Kernel::normalize (
            double epsilon = 1e-6 )
```

Here is the call graph for this function:

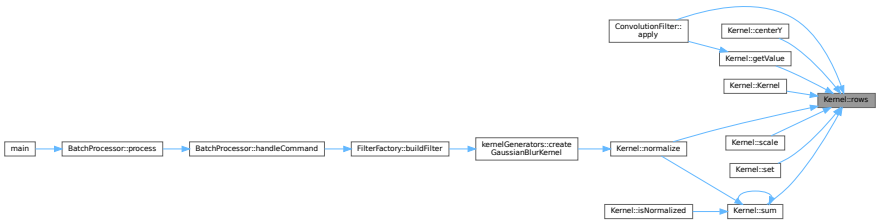


Here is the caller graph for this function:

6.13.2.10 rows()

```
int Kernel::rows ( ) const
```

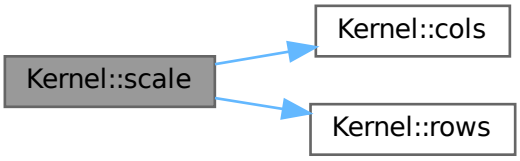
Here is the caller graph for this function:



6.13.2.11 scale()

```
void Kernel::scale (
            double factor )
```

Here is the call graph for this function:



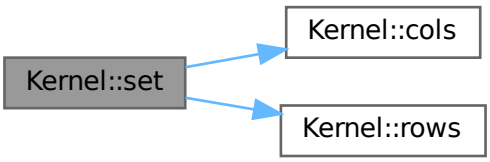
6.13.2.12 set() [1/2]

```
void Kernel::set (
            const KernelMatrix & values )
```

6.13.2.13 set() [2/2]

```
void Kernel::set (
            int row,
            int col,
            double value )
```

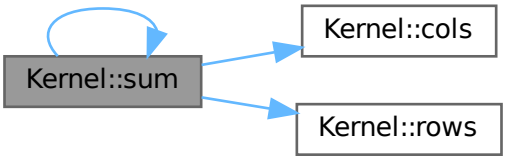
Here is the call graph for this function:



6.13.2.14 sum()

```
double Kernel::sum ( ) const
```

Here is the call graph for this function:



Here is the caller graph for this function:

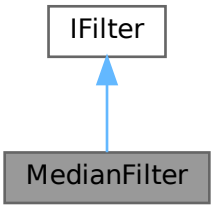
The documentation for this class was generated from the following files:

- include/types/Kernel.hpp
- src/types/Kernel.cpp

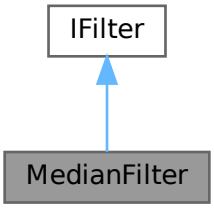
6.14 MedianFilter Class Reference

```
#include <MedianFilter.hpp>
```

Inheritance diagram for MedianFilter:



Collaboration diagram for MedianFilter:



Public Member Functions

- MedianFilter (unsigned kernelSize)
- void apply (Image &img) const override
- ~MedianFilter ()=default
- int absToRel (int coordinate, int size) const

Public Member Functions inherited from IFilter

- virtual ~IFilter ()=default

6.14.1 Constructor & Destructor Documentation

6.14.1.1 MedianFilter()

```
MedianFilter::MedianFilter (
            unsigned kernelSize )
```

6.14.1.2 ~MedianFilter()

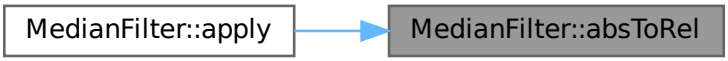
```
MedianFilter::~MedianFilter ( )  [default]
```

6.14.2 Member Function Documentation

6.14.2.1 absToRel()

```
int MedianFilter::absToRel (
            int coordinate,
            int size ) const
```

Here is the caller graph for this function:

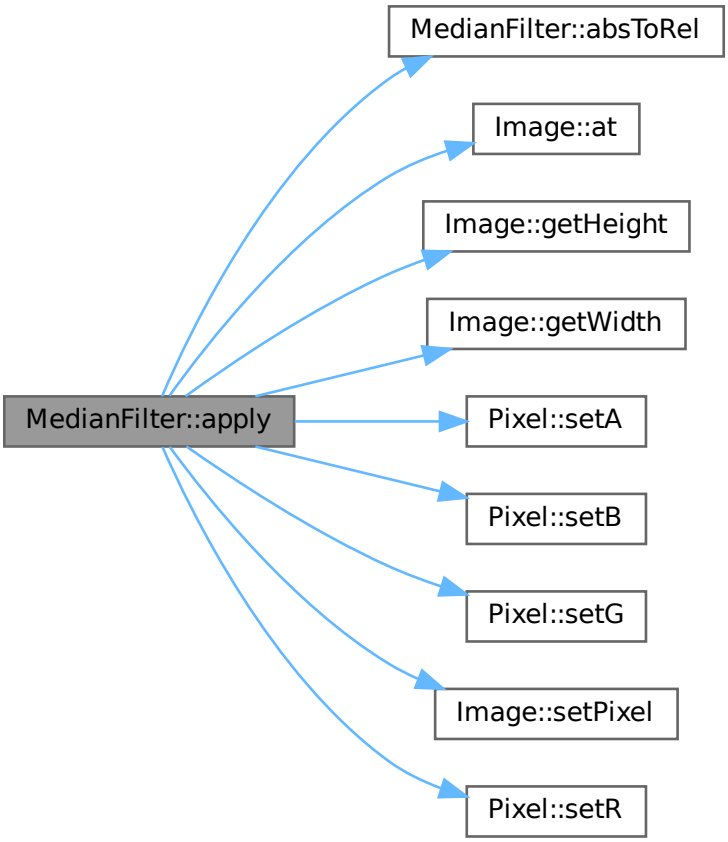


6.14.2.2 apply()

```
void MedianFilter::apply (
            Image & img ) const  [override], [virtual]
```

Implements IFilter.

Here is the call graph for this function:



The documentation for this class was generated from the following files:

- include/filters/MedianFilter.hpp
- src/filters/MedianFilter.cpp

6.15 Pixel Class Reference

```
#include <Pixel.hpp>
```

Public Member Functions

- Pixel (double r=0, double g=0, double b=0, double a=255)
- double getR () const
- double getG () const
- double getB () const
- double getA () const

- void setR (double value)
- void setG (double value)
- void setB (double value)
- void setA (double value)
- void setRGBA (double r, double g, double b, double a)
- void setPixel (const Pixel &pixel)
- Pixel operator* (double factor) const
- Pixel & operator+= (const Pixel &other)

6.15.1 Constructor & Destructor Documentation

6.15.1.1 Pixel()

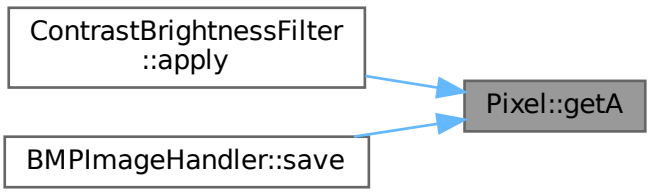
```
Pixel::Pixel (
            double r = 0,
            double g = 0,
            double b = 0,
            double a = 255 )  [inline]
```

6.15.2 Member Function Documentation

6.15.2.1 getA()

```
double Pixel::getA ( ) const  [inline]
```

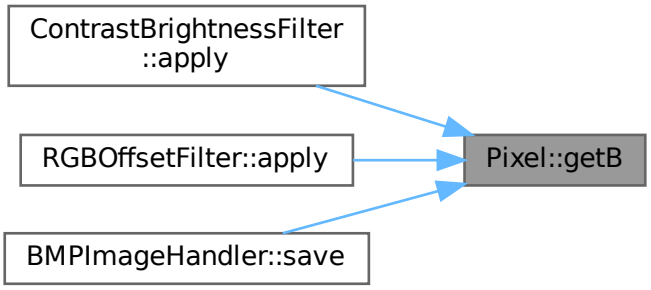
Here is the caller graph for this function:



6.15.2.2 getB()

```
double Pixel::getB ( ) const [inline]
```

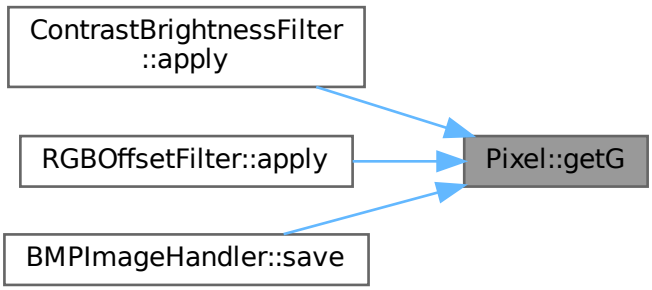
Here is the caller graph for this function:



6.15.2.3 getG()

```
double Pixel::getG ( ) const [inline]
```

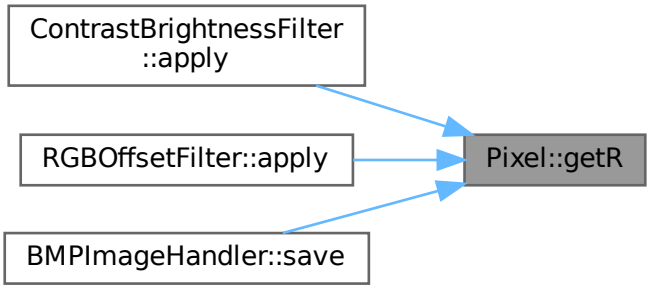
Here is the caller graph for this function:



6.15.2.4 getR()

```
double Pixel::getR ( ) const [inline]
```

Here is the caller graph for this function:



6.15.2.5 operator∗()

```
Pixel Pixel::operator* (
            double factor ) const  [inline]
```

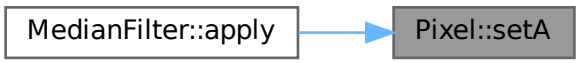
6.15.2.6 operator+=()

```
Pixel & Pixel::operator+= (
            const Pixel & other )  [inline]
```

6.15.2.7 setA()

```
void Pixel::setA (
            double value )  [inline]
```

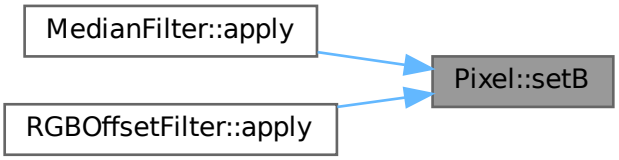
Here is the caller graph for this function:



6.15.2.8 setB()

```
void Pixel::setB (
            double value )  [inline]
```

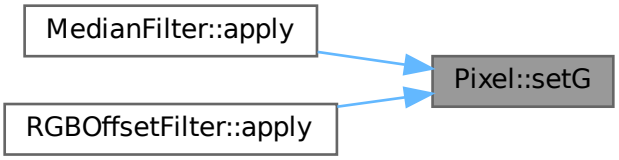
Here is the caller graph for this function:



6.15.2.9 setG()

```
void Pixel::setG (
            double value )  [inline]
```

Here is the caller graph for this function:



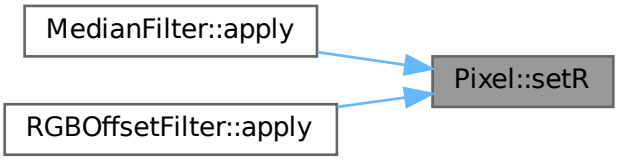
6.15.2.10 setPixel()

```
void Pixel::setPixel (
            const Pixel & pixel )  [inline]
```

6.15.2.11 setR()

```
void Pixel::setR (
            double value )  [inline]
```

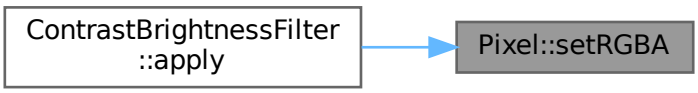
Here is the caller graph for this function:



6.15.2.12 setRGBA()

```
void Pixel::setRGBA (
            double r,
            double g,
            double b,
            double a )  [inline]
```

Here is the caller graph for this function:



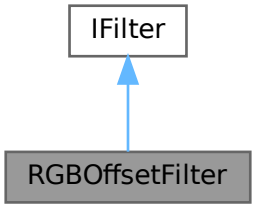
The documentation for this class was generated from the following file:

- include/types/Pixel.hpp

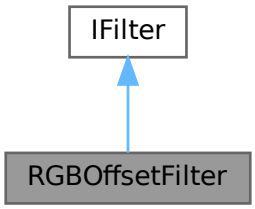
6.16 RGBOffsetFilter Class Reference

```
#include <RGBOffsetFilter.hpp>
```

Inheritance diagram for RGBOffsetFilter:



Collaboration diagram for RGBOffsetFilter:



Public Member Functions

- RGBOffsetFilter (double redOffset, double greenOffset, double blueOffset)
- void apply (Image &img) const override
- ~RGBOffsetFilter ()=default

Public Member Functions inherited from IFilter

- virtual ~IFilter ()=default

6.16.1 Constructor & Destructor Documentation

6.16.1.1 RGBOffsetFilter()

```
RGBOffsetFilter::RGBOffsetFilter (
            double redOffset,
            double greenOffset,
            double blueOffset )  [inline]
```

6.16.1.2 ~RGBOffsetFilter()

```
RGBOffsetFilter::~RGBOffsetFilter ( ) [default]
```

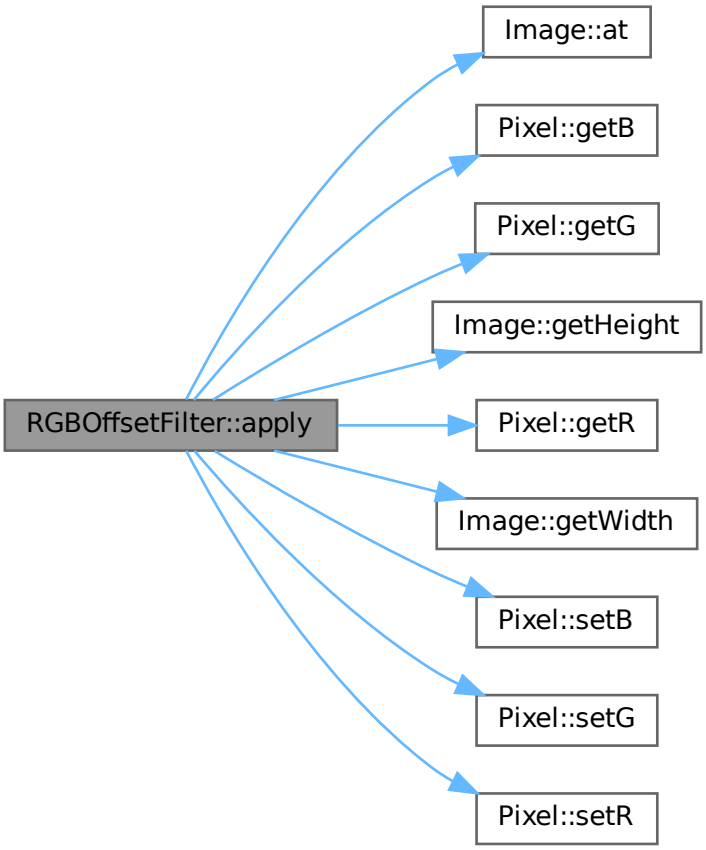
6.16.2 Member Function Documentation

6.16.2.1 apply()

```
void RGBOffsetFilter::apply (
            Image & img ) const [override], [virtual]
```

Implements IFilter.

Here is the call graph for this function:



The documentation for this class was generated from the following files:

- include/filters/RGBOffsetFilter.hpp
- src/filters/RGBOffsetFilter.cpp

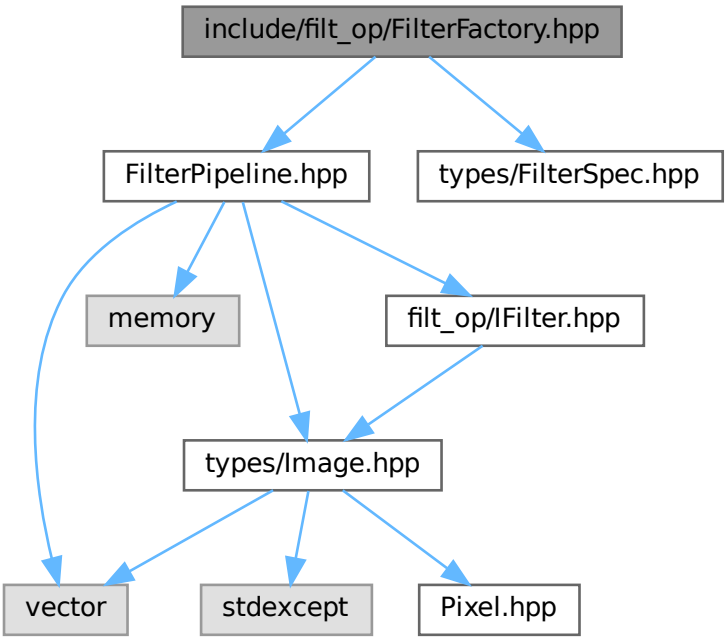
Chapter 7

File Documentation

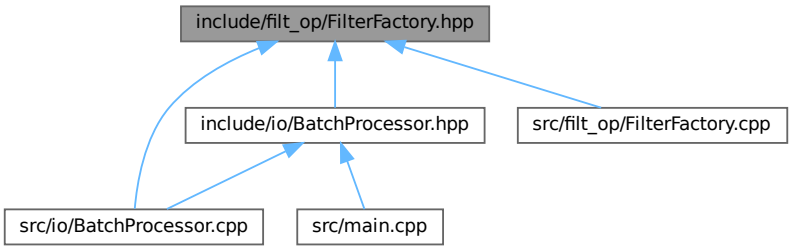
7.1 include/filt_op/FilterFactory.hpp File Reference

```
#include "FilterPipeline.hpp"
#include "types/FilterSpec.hpp"
```

Include dependency graph for FilterFactory.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class FilterFactory

7.2 FilterFactory.hpp

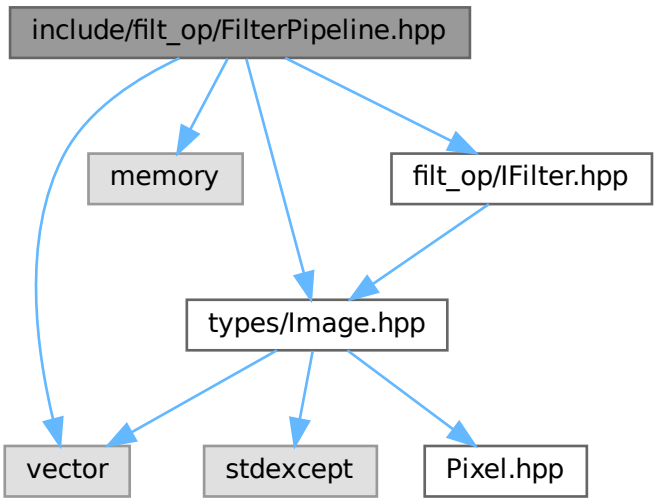
Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include "FilterPipeline.hpp"
00004 #include "types/FilterSpec.hpp"
00005
00006 class FilterFactory {
00007 public:
00008     static void buildFilter(FilterPipeline& pipeline, const FilterSpec& spec);
00009 };
```

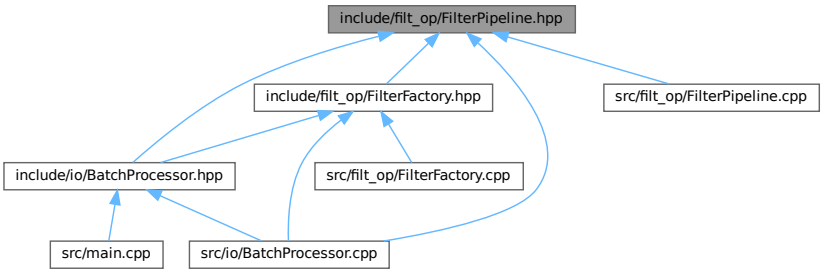
7.3 include/filt_op/FilterPipeline.hpp File Reference

```
#include <vector>
#include <memory>
#include "types/Image.hpp"
#include "filt_op/IFilter.hpp"
```

Include dependency graph for FilterPipeline.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class FilterPipeline

7.4 FilterPipeline.hpp

Go to the documentation of this file.

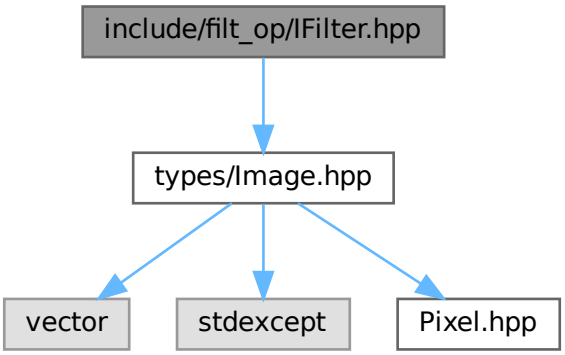
```
00001 #pragma once
00002
00003 #include <vector>
00004 #include <memory>
00005
00006 #include "types/Image.hpp"
00007 #include "filt_op/IFilter.hpp"
```

```
00008
00009 class FilterPipeline {
00010     std::vector<std::unique_ptr<IFilter» filters;
00011
00012 public:
00013     FilterPipeline() = default;
00014     ~FilterPipeline() = default;
00015
00016     void apply(Image& img);
00017     void add(std::unique_ptr<IFilter> filter);
00018     void clear();
00019 };
```

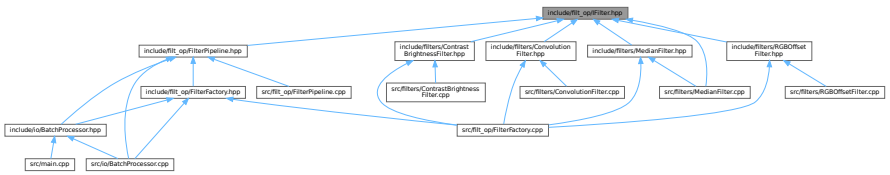
7.5 include/filt_op/IFilter.hpp File Reference

`#include "types/Image.hpp"`

Include dependency graph for IFilter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class IFilter

7.6 IFilter.hpp

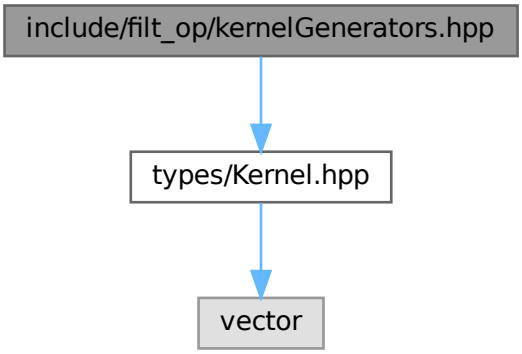
Go to the documentation of this file.

```
#pragma once

#include "types/Image.hpp"

// Interface for image filters used in FilterPipeline

class IFilter {
public:
    virtual ~IFilter() = default;
    virtual void apply(Image& img) const = 0;
};
```

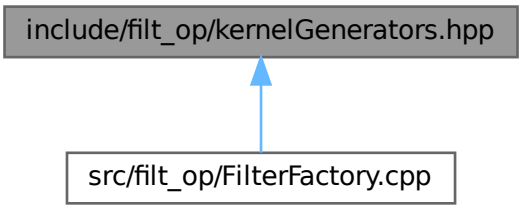
7.7 include/filt_op/kernelGenerators.hpp File Reference

`#include "types/Kernel.hpp"`

Include dependency graph for kernelGenerators.hpp:



This graph shows which files directly or indirectly include this file:



Namespaces

- namespace kernelGenerators

Functions

- Kernel kernelGenerators::createBoxBlurKernel (int size)
- Kernel kernelGenerators::createGaussianBlurKernel (int size, double sigma)

7.8 kernelGenerators.hpp

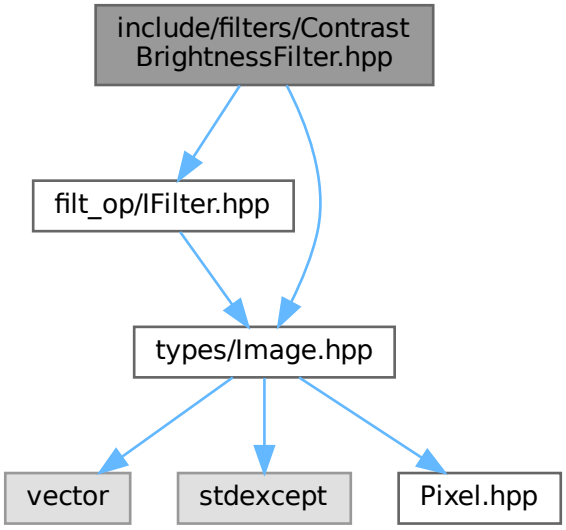
Go to the documentation of this file.

```
#pragma once

#include "types/Kernel.hpp"

namespace kernelGenerators {

Kernel createBoxBlurKernel(int size);

Kernel createGaussianBlurKernel(int size, double sigma);

}
```

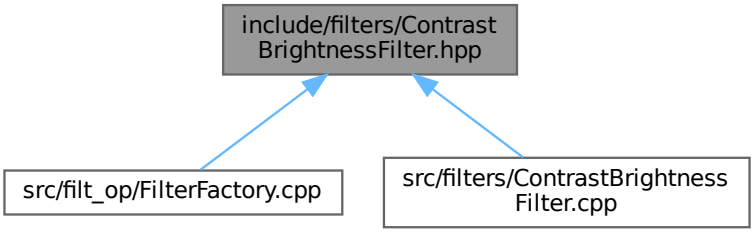
7.9 include/filters/ContrastBrightnessFilter.hpp File Reference

```
#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"
```

Include dependency graph for ContrastBrightnessFilter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class ContrastBrightnessFilter

7.10 ContrastBrightnessFilter.hpp

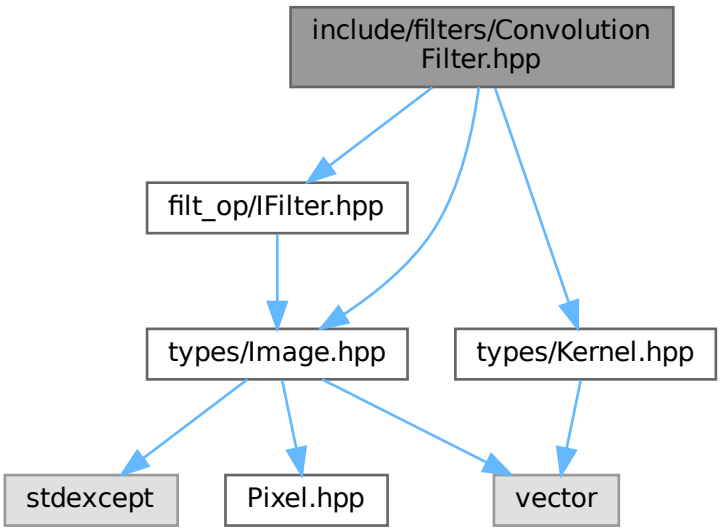
Go to the documentation of this file.

```
#pragma once

#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"


class ContrastBrightnessFilter : public IFilter {
    double contrastFactor = 1.0;
    double brightnessFactor = 0.0;
public:
   ContrastBrightnessFilter(double contrastFactor = 1.0, double brightnessFactor = 0.0) :
contrastFactor(contrastFactor), brightnessFactor(brightnessFactor) {}
   void apply(Image& img) const override;
   ~ContrastBrightnessFilter() = default;
};
```

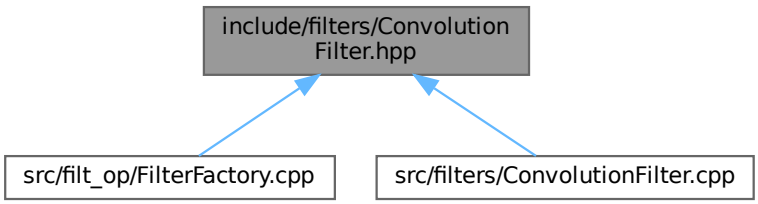
7.11 include/filters/ConvolutionFilter.hpp File Reference

```
#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"
#include "types/Kernel.hpp"
```

Include dependency graph for ConvolutionFilter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class ConvolutionFilter

7.12 ConvolutionFilter.hpp

Go to the documentation of this file.

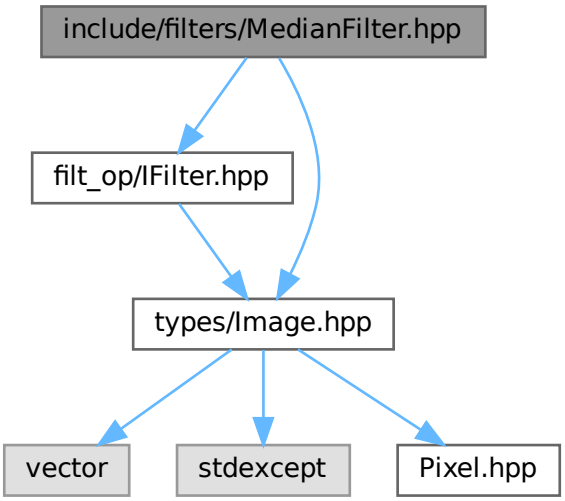
```
00001 #pragma once
00002
00003 #include "filt_op/IFilter.hpp"
00004 #include "types/Image.hpp"
00005 #include "types/Kernel.hpp"
00006
```

```
class ConvolutionFilter : public IFilter {
    Kernel kernel;

public:
    ConvolutionFilter(const Kernel& kernel) : kernel(kernel) {}
    ~ConvolutionFilter() = default;
    void apply(Image& img) const override;
};
```

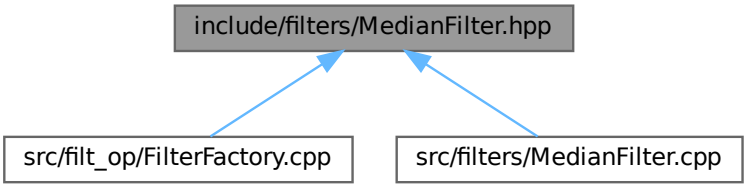
7.13 include/filters/MedianFilter.hpp File Reference

```
#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"
```

Include dependency graph for MedianFilter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class MedianFilter

7.14 MedianFilter.hpp

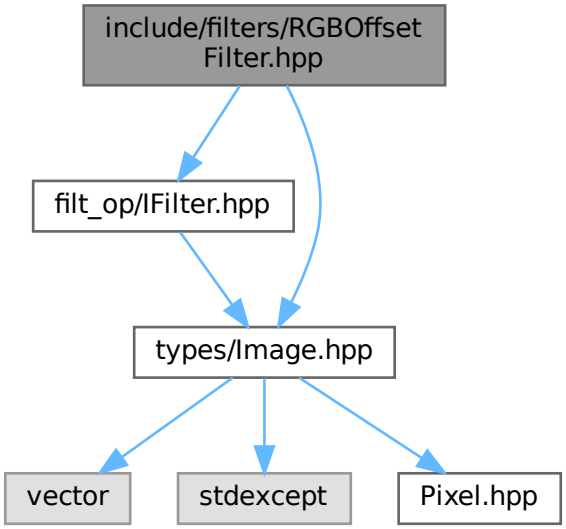
Go to the documentation of this file.

```
#pragma once

#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"

class MedianFilter : public IFilter {
    uint kernelSize; // the size of the kernel is kernelSize x kernelSize, it alway has to be odd!

public:
    MedianFilter(unsigned kernelSize);
    void apply(Image& img) const override;
    ~MedianFilter() = default;
    int absToRel(int coordinate, int size) const;

};
```

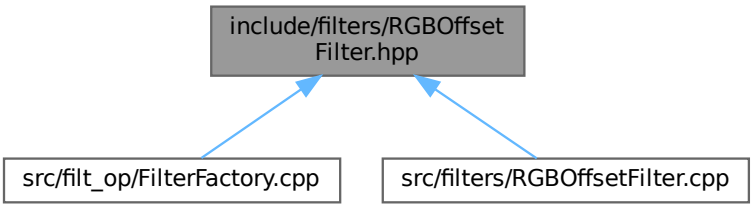
7.15 include/filters/RGBOffsetFilter.hpp File Reference

```
#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"
```

Include dependency graph for RGBOffsetFilter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class RGBOffsetFilter

7.16 RGBOffsetFilter.hpp

Go to the documentation of this file.

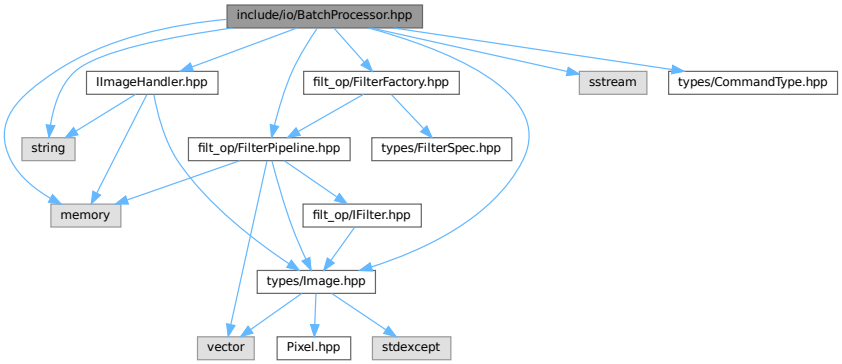
```
#pragma once

#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"


class RGBOffsetFilter : public IFilter {
    double redOffset;
    double greenOffset;
    double blueOffset;

public:
    RGBOffsetFilter(double redOffset, double greenOffset, double blueOffset) : redOffset(redOffset), greenOffset(greenOffset), blueOffset(blueOffset) {}
    void apply(Image& img) const override;
    ~RGBOffsetFilter() = default;
};
```

7.17 include/io/BatchProcessor.hpp File Reference

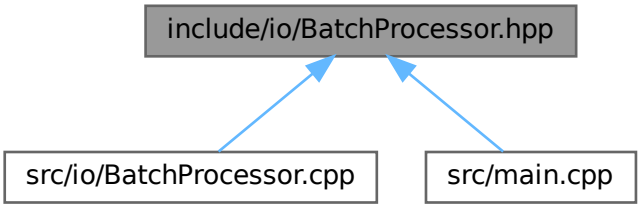
```
#include <string>
#include <memory>
#include <sstream>
#include "IImageHandler.hpp"
#include "filt_op/FilterPipeline.hpp"
#include "types/Image.hpp"
#include "filt_op/FilterFactory.hpp"
```

```
#include "types/CommandType.hpp"
```

Include dependency graph for BatchProcessor.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class BatchProcessor

7.18 BatchProcessor.hpp

Go to the documentation of this file.

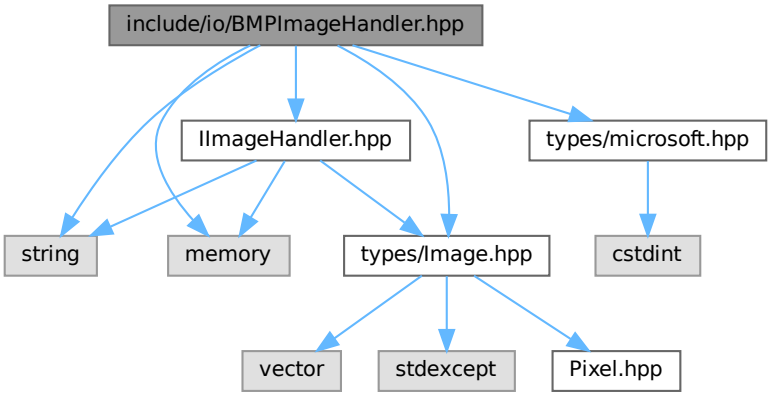
```
00001 #pragma once
00002
00003 #include <string>
00004 #include <memory>
00005 #include <sstream>
00006
00007 #include "IImageHandler.hpp"
00008 #include "filt_op/FilterPipeline.hpp"
00009 #include "types/Image.hpp"
00010 #include "filt_op/FilterFactory.hpp"
00011 #include "types/CommandType.hpp"
00012
00013
00014
00015
00016
```

```
class BatchProcessor{
    std::unique_ptr<Image> img;
    FilterPipeline pipeline;
    FilterFactory factory;
    FilterSpec spec;

    std::unique_ptr<IImageHandler> createHandlerFromExtension(const std::string& filename);
    CommandType getCommandType(const std::string& cmd);
public:
    BatchProcessor() = default;
    ~BatchProcessor() = default;

    void process(const std::string& filename);
    void handleCommand(const std::string& command, std::istringstream& args);
};
```

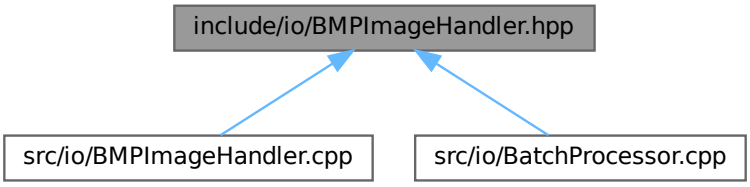
7.19 include/io/BMPImageHandler.hpp File Reference

```
#include <string>
#include <memory>
#include "IImageHandler.hpp"
#include "types/Image.hpp"
#include "types/microsoft.hpp"
```

Include dependency graph for BMPImageHandler.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class BMPImageHandler

7.20 BMPImageHandler.hpp

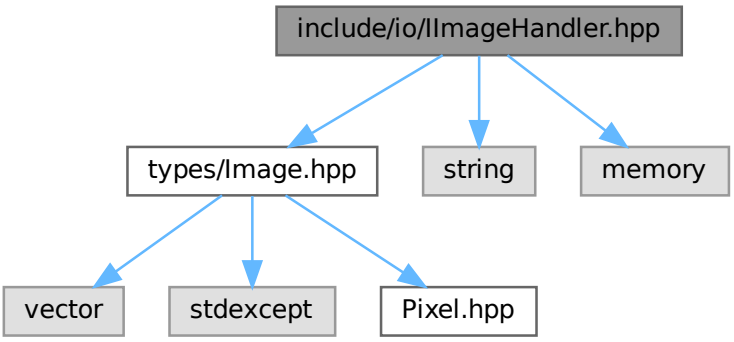
Go to the documentation of this file.

```
#pragma once

#include <string>
#include <memory>

#include "IImageHandler.hpp"
#include "types/Image.hpp"
#include "types/microsoft.hpp"


class BMPImageHandler : public IImageHandler {
public:
    BMPImageHandler() = default;
    ~BMPImageHandler() override;

    std::unique_ptr<Image> load(const std::string& filename) const override;
    void save(const std::string& filename, const Image& image) const override;

    void flipVertical(Image& img, BITMAPINFOHEADER& header) const;
};
```

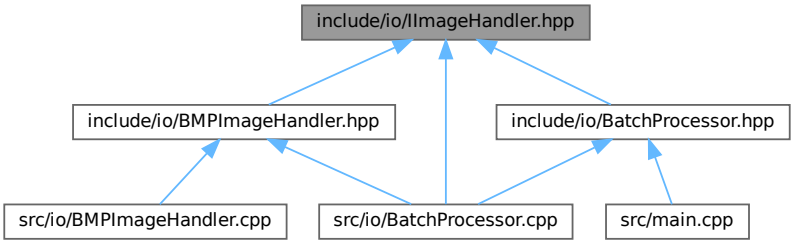
7.21 include/io/IImageHandler.hpp File Reference

```
#include "types/Image.hpp"
#include <string>
#include <memory>
```

Include dependency graph for IImageHandler.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class IImageHandler

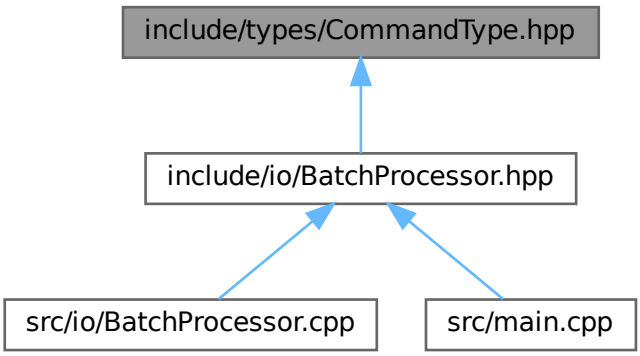
7.22 IImageHandler.hpp

Go to the documentation of this file.

```
#pragma once
#include "types/Image.hpp"
#include <string>
#include <memory>

class IImageHandler {
public:
    virtual ~IImageHandler() = default;
    virtual std::unique_ptr<Image> load(const std::string& filename) const = 0;
    virtual void save(const std::string& filename, const Image& image) const = 0;
};
```

7.23 include/types/CommandType.hpp File Reference

This graph shows which files directly or indirectly include this file:



Enumerations

- enum class CommandType {
  INPUT , OUTPUT , CONTRAST , BRIGHTNESS ,
  BLUR , GAUSSIAN_BLUR , MEDIAN , RGB_OFFSET ,
  UNKNOWN }

7.23.1 Enumeration Type Documentation

7.23.1.1 CommandType

```
enum class CommandType  [strong]
```

Enumerator

INPUT	
OUTPUT	
CONTRAST	
BRIGHTNESS	
BLUR	
GAUSSIAN_BLUR	
MEDIAN	
RGB_OFFSET	
UNKNOWN	

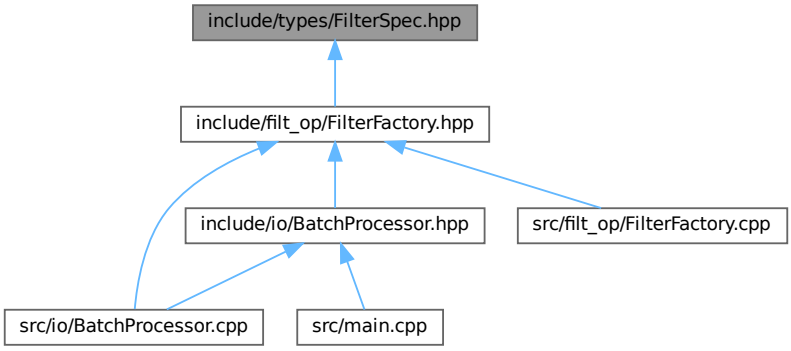
7.24 CommandType.hpp

Go to the documentation of this file.

```
#pragma once

enum class CommandType {
    INPUT,
    OUTPUT,
    CONTRAST,
    BRIGHTNESS,
    BLUR,
    GAUSSIAN_BLUR,
    MEDIAN,
    RGB_OFFSET,
    UNKNOWN,
};
```

7.25 include/types/FilterSpec.hpp File Reference

This graph shows which files directly or indirectly include this file:



Classes

- class FilterSpec

7.26 FilterSpec.hpp

Go to the documentation of this file.

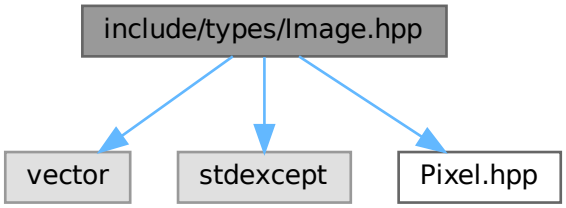
```
#pragma once

class FilterSpec {
public:
    FilterSpec() {
        clear();
    }

    ~FilterSpec() = default;

    bool hasContrast = false;
    double contrastFactor = 0;

    bool hasBrightness = false;
    double brightnessFactor = 0;

    bool hasBlur = false;
    unsigned blur = 0; // must be odd

    bool hasGaussian = false;
    unsigned gaussianBlurSize = 0; // must be odd
    double gaussianBlurSigma = 1.0;

    bool hasMedian = false;
    uint medianKernelSize = 0; // must be odd

    bool hasRGBOffset = false;
    double rOffset = 0;
    double gOffset = 0;
    double bOffset = 0;

    void clear() {
        hasContrast = false;
        contrastFactor = 0;

        hasBrightness = false;
        brightnessFactor = 0;

```

```
        hasBlur = false;
        blur = 0;

        hasGaussian = false;
        gaussianBlurSize = 0;
        gaussianBlurSigma = 1.0;

        hasMedian = false;
        medianKernelSize = 0;

        hasRGBOffset = false;
        rOffset = gOffset = bOffset = 0;
    }
};
```

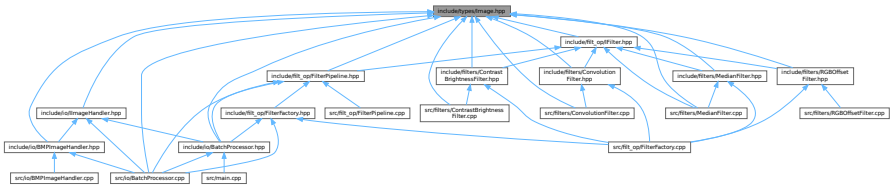
7.27 include/types/Image.hpp File Reference

```
#include <vector>
#include <stdexcept>
#include "Pixel.hpp"
```

Include dependency graph for Image.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Image

7.28 Image.hpp

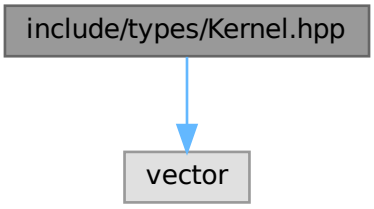
Go to the documentation of this file.

```
#pragma once

#include <vector>
#include <stdexcept>

#include "Pixel.hpp"

class Image {
    std::vector<std::vector<Pixel» pixels; // pixels[row][col] - when writing pixels[row] it gives you a row
    int width, height;
public:
    Image() = default;

    Image(int width, int height) : width(width), height(height) {
        pixels.resize(height, std::vector<Pixel>(width));
    }

    const Pixel& at(int row, int col) const { // works on constant objects
        if (row < 0 || row >= height || col < 0 || col >= width)
            throw std::out_of_range("Pixel coordinates out of range");
        return pixels[row][col];
    }

    Pixel& at(int row, int col) {
        if (row < 0 || row >= height || col < 0 || col >= width)
            throw std::out_of_range("Pixel coordinates out of range");
        return pixels[row][col];
    }

    void resize(int width, int height) {
        this->width = width;
        this->height = height;
        pixels.resize(height, std::vector<Pixel>(width));
    }

    void setPixel(int row, int col, const Pixel& pixel) {
        if (row < 0 || row >= height || col < 0 || col >= width)
            throw std::out_of_range("Pixel coordinates out of range");
        pixels[row][col] = pixel;
    }

    int getWidth() const { return width; }

    int getHeight() const { return height; }
};
```

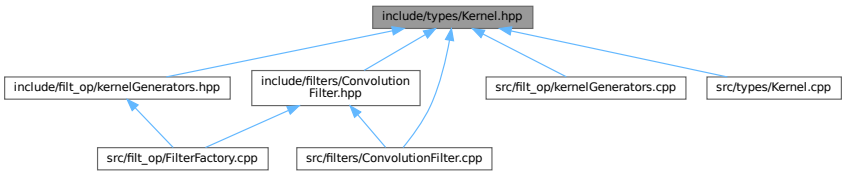
7.29 include/types/Kernel.hpp File Reference

`#include <vector>`

Include dependency graph for Kernel.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Kernel

Typedefs

- using KernelMatrix = std::vector< std::vector< double > >

7.29.1 Typedef Documentation

7.29.1.1 KernelMatrix

```
using KernelMatrix = std::vector<std::vector<double> >
```

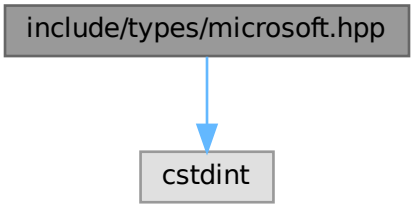
7.30 Kernel.hpp

Go to the documentation of this file.

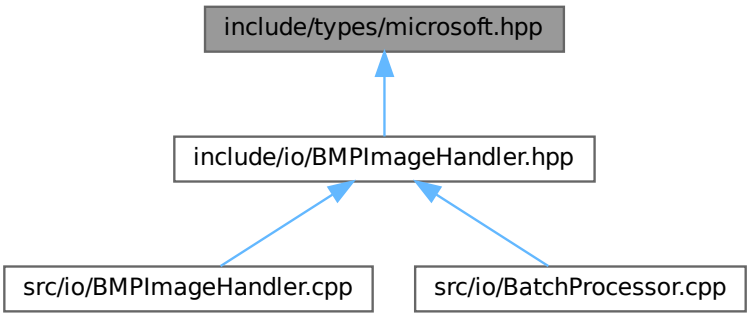
```
#pragma once

#include <vector>

using KernelMatrix = std::vector<std::vector<double»;

class Kernel {
    KernelMatrix kernelValues;

public:
    Kernel() = default; // "=default" should be always in .hpp
    Kernel(KernelMatrix values);

    double getValue(int row, int col) const;
    void set(int row, int col, double value);
    void set(const KernelMatrix& values);
    const KernelMatrix& getValues() const;
    KernelMatrix& getValues();

    int centerX() const;
    int centerY() const;
    int rows() const; // height
    int cols() const; // width
    int absToRel(int coordinate, int size) const; // convert absolute coordinates to relative coordinates to the center of the kernel

    double sum() const;
    void normalize(double epsilon = 1e-6);
    bool isNormalized(double epsilon = 1e-6) const;
    void scale(double factor);
};
```

7.31 include/types/microsoft.hpp File Reference

`#include <cstdint>`
Include dependency graph for microsoft.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct BITMAPFILEHEADER
- struct BITMAPINFOHEADER

7.32 microsoft.hpp

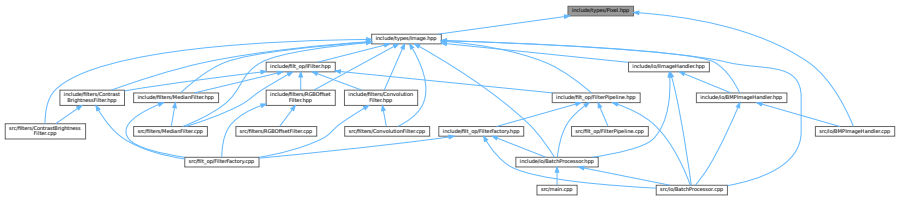
Go to the documentation of this file.

```
#pragma once

#include <cstdint>

// Nedded, if missing, the compiler may add padding bytes to the structures, and that messes up the reading in
#pragma pack(push, 1)
struct BITMAPFILEHEADER {
```

```
    uint16_t fileType;
    uint32_t fileSize;
    uint16_t reserved1;
    uint16_t reserved2;
    uint32_t offsetData;
};

struct BITMAPINFOHEADER {
    uint32_t size;
    int32_t width;
    int32_t height;
    uint16_t planes;
    uint16_t bitCount;
    uint32_t compression;
    uint32_t sizeImage;
    int32_t xPixelsPerMeter;
    int32_t yPixelsPerMeter;
    uint32_t colorsUsed;
    uint32_t importantColors;
};
#pragma pack(pop)
```

7.33 include/types/Pixel.hpp File Reference

This graph shows which files directly or indirectly include this file:



Classes

- class Pixel

7.34 Pixel.hpp

Go to the documentation of this file.

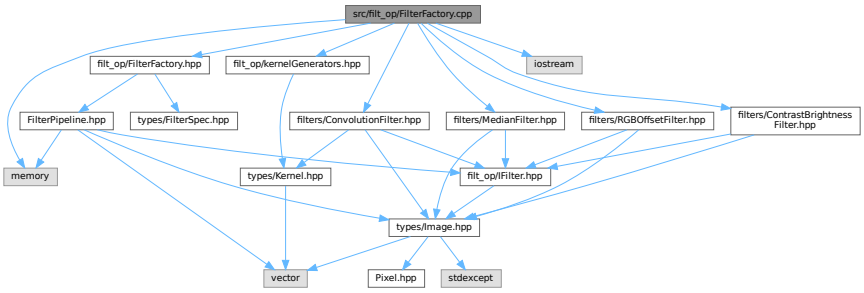
```
#pragma once

class Pixel {
    double r, g, b, a;

public:
    Pixel(double r = 0, double g = 0, double b = 0, double a = 255)
        : r(r), g(g), b(b), a(a) {}

    double getR() const { return r; }
    double getG() const { return g; }
    double getB() const { return b; }
    double getA() const { return a; }


    void setR(double value) { r = value; }
    void setG(double value) { g = value; }
    void setB(double value) { b = value; }
    void setA(double value) { a = value; }
    void setRGBA(double r, double g, double b, double a) {
        this->r = r;
        this->g = g;
        this->b = b;
        this->a = a;
```

```
    }
    void setPixel(const Pixel& pixel) {
        r = pixel.r;
        g = pixel.g;
        b = pixel.b;
        a = pixel.a;
    }

    Pixel operator*(double factor) const{
        return Pixel(r * factor, g * factor, b * factor, a);
    }

    Pixel& operator+=(const Pixel& other) {
        r += other.r;
        g += other.g;
        b += other.b;
        a = (a + other.a) / 2; // average alpha
        return *this;
    }
};
```

7.35 src/filt_op/FilterFactory.cpp File Reference

```
#include <memory>
#include <iostream>
#include "filt_op/kernelGenerators.hpp"
#include "filt_op/FilterFactory.hpp"
#include "filters/ContrastBrightnessFilter.hpp"
#include "filters/ConvolutionFilter.hpp"
#include "filters/MedianFilter.hpp"
#include "filters/RGBOffsetFilter.hpp"
```

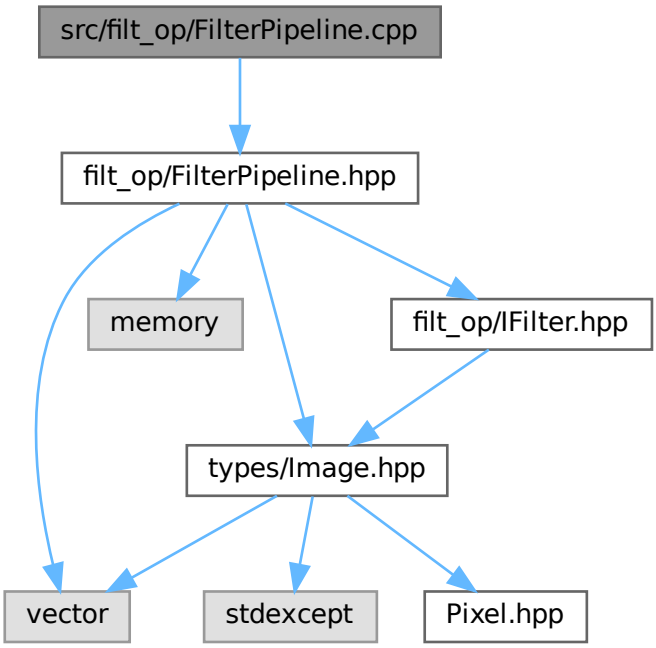
Include dependency graph for FilterFactory.cpp:



7.36 src/filt_op/FilterPipeline.cpp File Reference

```
#include "filt_op/FilterPipeline.hpp"
```

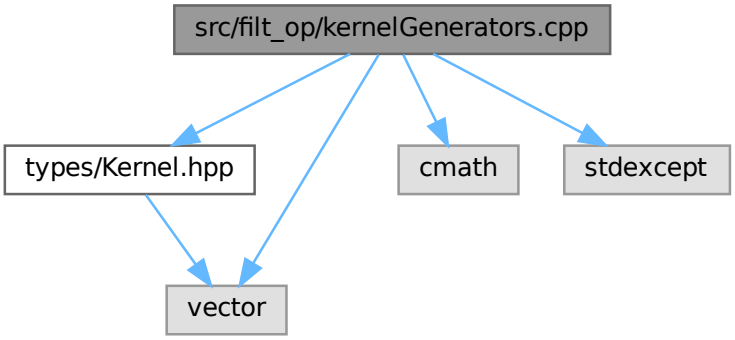
Include dependency graph for FilterPipeline.cpp:



7.37 src/filt_op/kernelGenerators.cpp File Reference

```
#include "types/Kernel.hpp"
#include <cmath>
#include <stdexcept>
#include <vector>
```

Include dependency graph for kernelGenerators.cpp:



Namespaces

- namespace kernelGenerators

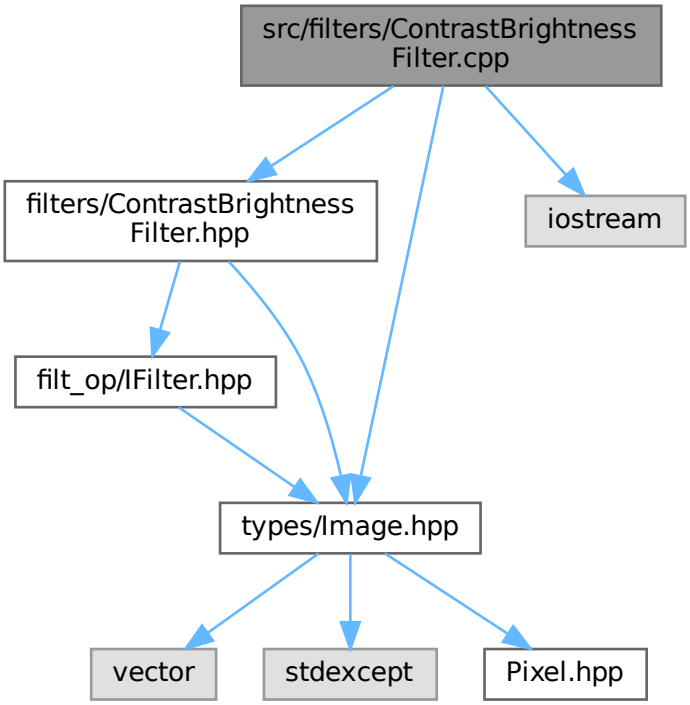
Functions

- Kernel kernelGenerators::createBoxBlurKernel (int size)
- Kernel kernelGenerators::createGaussianBlurKernel (int size, double sigma)

7.38 src/filters/ContrastBrightnessFilter.cpp File Reference

```
#include "filters/ContrastBrightnessFilter.hpp"
#include "types/Image.hpp"
#include <iostream>
```

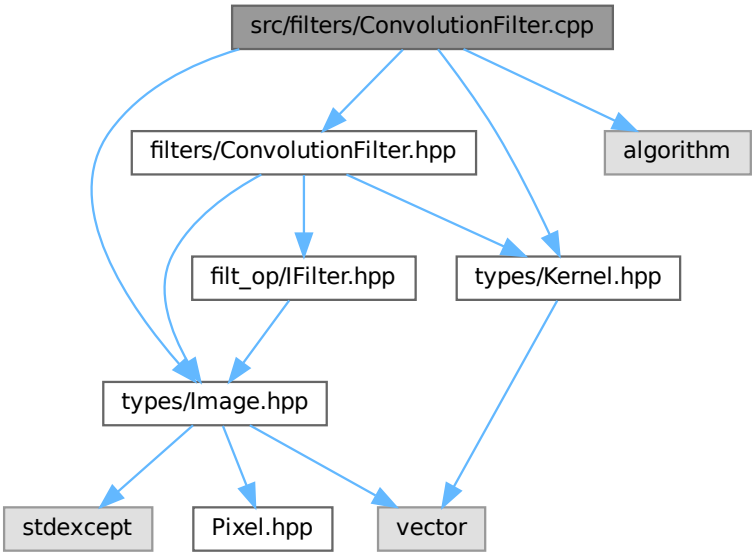
Include dependency graph for ContrastBrightnessFilter.cpp:



7.39 src/filters/ConvolutionFilter.cpp File Reference

```
#include "filters/ConvolutionFilter.hpp"
#include "types/Image.hpp"
#include "types/Kernel.hpp"
#include <algorithm>
```

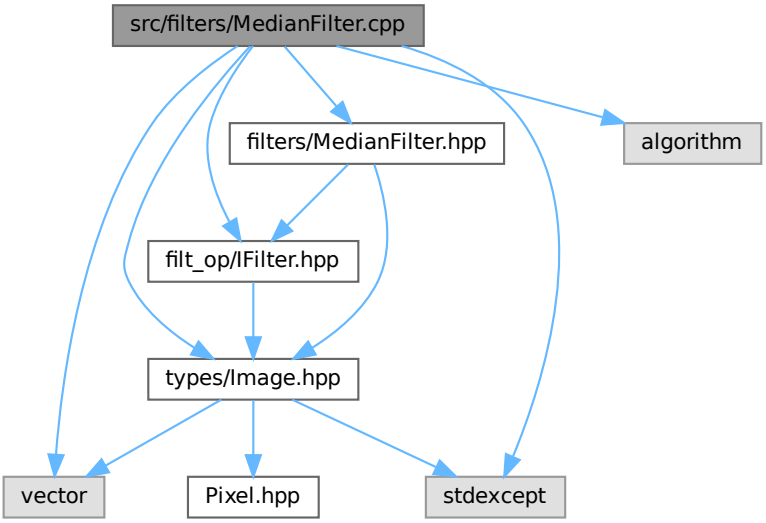
Include dependency graph for ConvolutionFilter.cpp:



7.40 src/filters/MedianFilter.cpp File Reference

```
#include "filters/MedianFilter.hpp"
#include "filt_op/IFilter.hpp"
#include "types/Image.hpp"
#include <algorithm>
#include <vector>
#include <stdexcept>
```

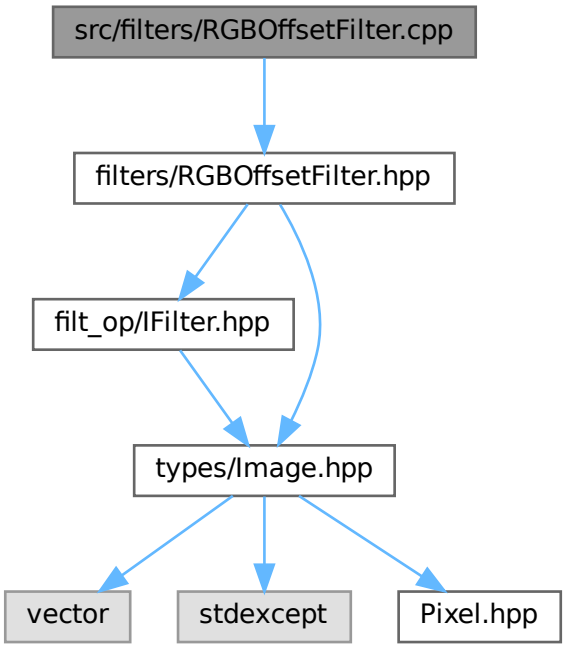
Include dependency graph for MedianFilter.cpp:



7.41 src/filters/RGBOffsetFilter.cpp File Reference

```
#include "filters/RGBOffsetFilter.hpp"
```

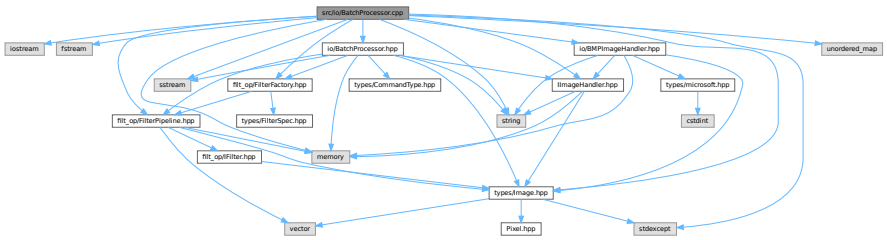
Include dependency graph for RGBOffsetFilter.cpp:



7.42 src/io/BatchProcessor.cpp File Reference

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
#include <stdexcept>
#include <memory>
#include <unordered_map>
#include "io/BatchProcessor.hpp"
#include "io/IImageHandler.hpp"
#include "filt_op/FilterPipeline.hpp"
#include "types/Image.hpp"
#include "filt_op/FilterFactory.hpp"
#include "io/BMPImageHandler.hpp"
```

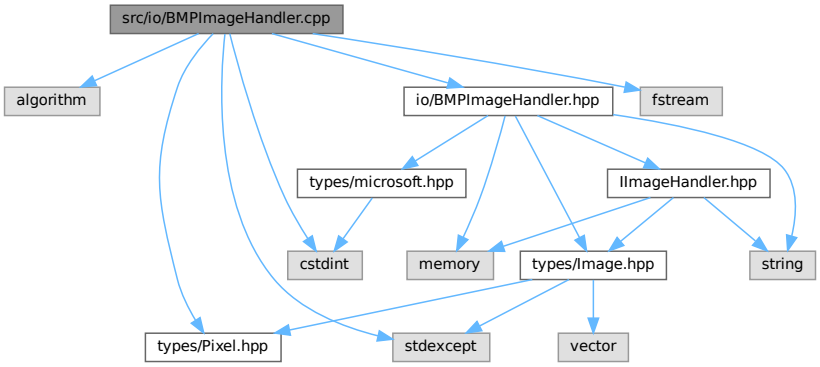
Include dependency graph for BatchProcessor.cpp:



7.43 src/io/BMPImageHandler.cpp File Reference

```
#include <algorithm>
#include "types/Pixel.hpp"
#include <stdexcept>
#include "io/BMPImageHandler.hpp"
#include <fstream>
#include <cstdint>
```

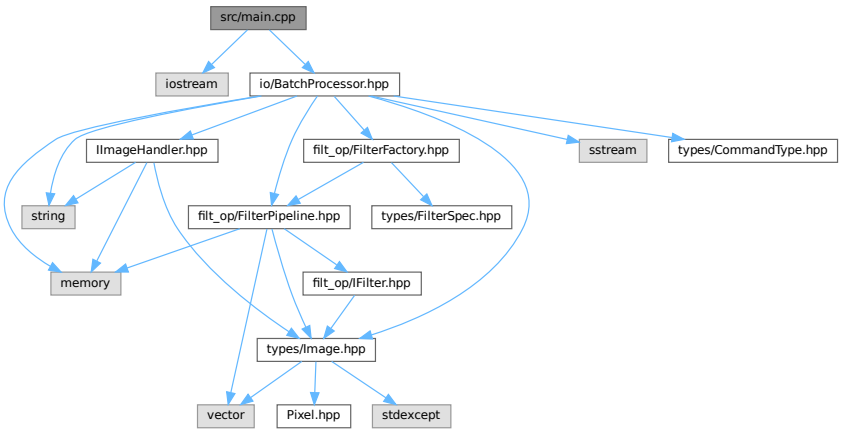
Include dependency graph for BMPImageHandler.cpp:



7.44 src/main.cpp File Reference

```
#include <iostream>
#include "io/BatchProcessor.hpp"
```

Include dependency graph for main.cpp:



Functions

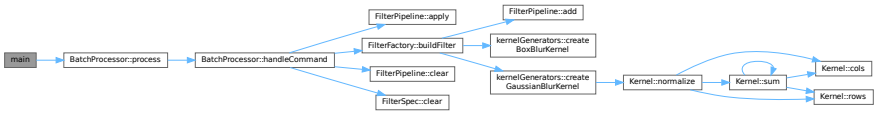
- int main (int argc, char *argv[])

7.44.1 Function Documentation

7.44.1.1 main()

```
int main (
            int argc,
            char * argv[] )
```

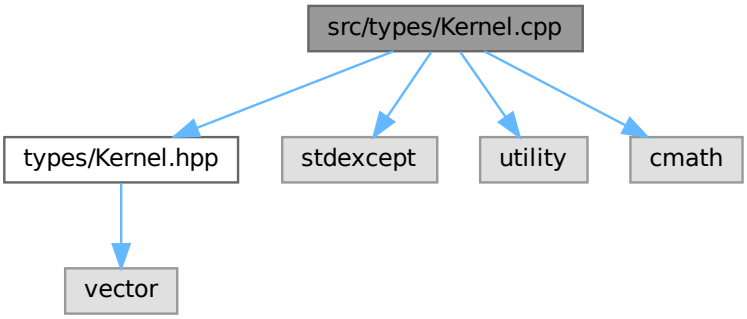
Here is the call graph for this function:



7.45 src/types/Kernel.cpp File Reference

```
#include "types/Kernel.hpp"
#include <stdexcept>
#include <utility>
#include <cmath>
```

Include dependency graph for Kernel.cpp:



Index

~BMPImageHandler
    BMPImageHandler, 16
~BatchProcessor
    BatchProcessor, 11
~ContrastBrightnessFilter
    ContrastBrightnessFilter, 19
~ConvolutionFilter
    ConvolutionFilter, 21
~FilterPipeline
    FilterPipeline, 24
~FilterSpec
    FilterSpec, 25
~IFilter
    IFilter, 28
~IImageHandler
    IImageHandler, 29
~MedianFilter
    MedianFilter, 40
~RGBOffsetFilter
    RGBOffsetFilter, 48

absToRel
    Kernel, 34
    MedianFilter, 41
add
    FilterPipeline, 24
apply
    ContrastBrightnessFilter, 20
    ConvolutionFilter, 22
    FilterPipeline, 24
    IFilter, 28
    MedianFilter, 41
    RGBOffsetFilter, 49
at
    Image, 30

BatchProcessor, 11
    ~BatchProcessor, 11
    BatchProcessor, 11
    handleCommand, 12
    process, 12
bitCount
    BITMAPINFOHEADER, 14
BITMAPFILEHEADER, 13
    fileSize, 13
    fileType, 13
    offsetData, 13
    reserved1, 13
    reserved2, 13
BITMAPINFOHEADER, 14
    bitCount, 14
    colorsUsed, 14
    compression, 14
    height, 14
    importantColors, 14
    planes, 14
    size, 14
    sizeImage, 15
    width, 15
    xPixelsPerMeter, 15
    yPixelsPerMeter, 15
BLUR
    CommandType.hpp, 66
blur
    FilterSpec, 26
BMPImageHandler, 15
    ~BMPImageHandler, 16
    BMPImageHandler, 16
    flipVertical, 17
    load, 17
    save, 17
bOffset
    FilterSpec, 26
BRIGHTNESS
    CommandType.hpp, 66
brightnessFactor
    FilterSpec, 26
buildFilter
    FilterFactory, 23

centerX
    Kernel, 34
centerY
    Kernel, 34
clear
    FilterPipeline, 24
    FilterSpec, 26
colorsUsed
    BITMAPINFOHEADER, 14
cols
    Kernel, 35
CommandType
    CommandType.hpp, 66
CommandType.hpp
    BLUR, 66
    BRIGHTNESS, 66
    CommandType, 66
    CONTRAST, 66
    GAUSSIAN_BLUR, 66
    INPUT, 66

MEDIAN, 66
OUTPUT, 66
RGB_OFFSET, 66
UNKNOWN, 66
compression
BITMAPINFOHEADER, 14
CONTRAST
CommandType.hpp, 66
ContrastBrightnessFilter, 18
~ContrastBrightnessFilter, 19
apply, 20
ContrastBrightnessFilter, 19
contrastFactor
FilterSpec, 26
ConvolutionFilter, 21
~ConvolutionFilter, 21
apply, 22
ConvolutionFilter, 21
createBoxBlurKernel
kernelGenerators, 9
createGaussianBlurKernel
kernelGenerators, 9

fileSize
BITMAPFILEHEADER, 13
fileType
BITMAPFILEHEADER, 13
FilterFactory, 22
buildFilter, 23
FilterPipeline, 23
~FilterPipeline, 24
add, 24
apply, 24
clear, 24
FilterPipeline, 24
FilterSpec, 25
~FilterSpec, 25
blur, 26
bOffset, 26
brightnessFactor, 26
clear, 26
contrastFactor, 26
FilterSpec, 25
gaussianBlurSigma, 26
gaussianBlurSize, 26
gOffset, 26
hasBlur, 26
hasBrightness, 27
hasContrast, 27
hasGaussian, 27
hasMedian, 27
hasRGBOffset, 27
medianKernelSize, 27
rOffset, 27
flipVertical
BMPImageHandler, 17

GAUSSIAN_BLUR
CommandType.hpp, 66
gaussianBlurSigma
FilterSpec, 26
gaussianBlurSize
FilterSpec, 26
getA
Pixel, 43
getB
Pixel, 43
getG
Pixel, 44
getHeight
Image, 31
getR
Pixel, 44
getValue
Kernel, 35
getValues
Kernel, 36
getWidth
Image, 31
gOffset
FilterSpec, 26

handleCommand
BatchProcessor, 12
hasBlur
FilterSpec, 26
hasBrightness
FilterSpec, 27
hasContrast
FilterSpec, 27
hasGaussian
FilterSpec, 27
hasMedian
FilterSpec, 27
hasRGBOffset
FilterSpec, 27
height
BITMAPINFOHEADER, 14

IFilter, 28
~IFilter, 28
apply, 28
IImageHandler, 29
~IImageHandler, 29
load, 29
save, 29
Image, 30
at, 30
getHeight, 31
getWidth, 31
Image, 30
resize, 32
setPixel, 32
importantColors
BITMAPINFOHEADER, 14
include/filt_op/FilterFactory.hpp, 51, 52
include/filt_op/FilterPipeline.hpp, 52, 53
include/filt_op/IFilter.hpp, 54, 55

include/filt_op/kernelGenerators.hpp, 55, 56
include/filters/ContrastBrightnessFilter.hpp, 56, 57
include/filters/ConvolutionFilter.hpp, 57, 58
include/filters/MedianFilter.hpp, 59, 60
include/filters/RGBOffsetFilter.hpp, 60, 61
include/io/BatchProcessor.hpp, 61, 62
include/io/BMPImageHandler.hpp, 63, 64
include/io/IImageHandler.hpp, 64, 65
include/types/CommandType.hpp, 65, 66
include/types/FilterSpec.hpp, 67
include/types/Image.hpp, 68, 69
include/types/Kernel.hpp, 69, 70
include/types/microsoft.hpp, 71
include/types/Pixel.hpp, 72
INPUT
    CommandType.hpp, 66
isNormalized
    Kernel, 36

Kernel, 33
    absToRel, 34
    centerX, 34
    centerY, 34
    cols, 35
    getValue, 35
    getValues, 36
    isNormalized, 36
    Kernel, 33
    normalize, 36
    rows, 37
    scale, 37
    set, 38
    sum, 38
Kernel.hpp
    KernelMatrix, 70
kernelGenerators, 9
    createBoxBlurKernel, 9
    createGaussianBlurKernel, 9
KernelMatrix
    Kernel.hpp, 70

load
    BMPImageHandler, 17
    IImageHandler, 29

main
    main.cpp, 81
main.cpp
    main, 81
MEDIAN
    CommandType.hpp, 66
MedianFilter, 39
    ∼MedianFilter, 40
    absToRel, 41
    apply, 41
    MedianFilter, 40
medianKernelSize
    FilterSpec, 27

normalize
    Kernel, 36

offsetData
    BITMAPFILEHEADER, 13
operator+=
    Pixel, 45
operator∗
    Pixel, 45
OUTPUT
    CommandType.hpp, 66

Pixel, 42
    getA, 43
    getB, 43
    getG, 44
    getR, 44
    operator+=, 45
    operator∗, 45
    Pixel, 43
    setA, 45
    setB, 45
    setG, 46
    setPixel, 46
    setR, 46
    setRGBA, 47
planes
    BITMAPINFOHEADER, 14
process
    BatchProcessor, 12

reserved1
    BITMAPFILEHEADER, 13
reserved2
    BITMAPFILEHEADER, 13
resize
    Image, 32
RGB_OFFSET
    CommandType.hpp, 66
RGBOffsetFilter, 48
    ∼RGBOffsetFilter, 48
    apply, 49
    RGBOffsetFilter, 48
rOffset
    FilterSpec, 27
rows
    Kernel, 37

save
    BMPImageHandler, 17
    IImageHandler, 29
scale
    Kernel, 37
set
    Kernel, 38
setA
    Pixel, 45
setB
    Pixel, 45

setG
Pixel, 46
setPixel
Image, 32
Pixel, 46
setR
Pixel, 46
setRGBA
Pixel, 47
size
BITMAPINFOHEADER, 14
sizeImage
BITMAPINFOHEADER, 15
src/filt_op/FilterFactory.cpp, 73
src/filt_op/FilterPipeline.cpp, 74
src/filt_op/kernelGenerators.cpp, 74
src/filters/ContrastBrightnessFilter.cpp, 75
src/filters/ConvolutionFilter.cpp, 76
src/filters/MedianFilter.cpp, 77
src/filters/RGBOffsetFilter.cpp, 78
src/io/BatchProcessor.cpp, 79
src/io/BMPImageHandler.cpp, 80
src/main.cpp, 80
src/types/Kernel.cpp, 81
sum
Kernel, 38

UNKNOWN
CommandType.hpp, 66

width
BITMAPINFOHEADER, 15

xPixelsPerMeter
BITMAPINFOHEADER, 15

yPixelsPerMeter
BITMAPINFOHEADER, 15